



DEPARTMENT VISION

To endorse knowledge and invention that incorporates engineering and life sciences for the progression of human health. To build a centre of academic excellence in biomedical engineering for betterment in healthcare through industrial oriented teaching and research contributions. The Biomedical engineering department is to become a centre of excellence to deliver a quality health service for the success of the country through education and research.

DEPARTMENT MISSION

- To train the students as knowledge-based engineers by innovative thinking to meet the challenges in the field of medical and electronics engineering for healthier India.
- To encourage faculty members and students to detonate their ingenuity to cultivate innovative products by employing modern technologies to assist the society.
- To teach the students on the industrial need in biomedical engineering through significant training and value-added courses.
- To bring out technically passionate engineers through innovative teaching practices.
- To educate students to realize that human body is a combined system, through computable engineering investigation and customizing to design better therapeutic policies, devices, and diagnostics.

PROGRAM OUTCOMES (POs)

1	Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2	Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3	Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental Considerations.
4	Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5	Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6	The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7	Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

9	Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11	Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to ones own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12	Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

PEO 1	To educate students with quality foundation in mathematical, scientific and engineering basics which are essential to communicate, analyse and solve engineering complication in healthcare sector.
PEO 2	To exhibit leadership, decision making with ethical responsibilities, utility and communicate efficiently in multidisciplinary fields of Engineering.
PEO 3	To teach students with assistances for research, design and development of biomedical devices and associated integrated systems for the benefit of human society.
PEO 4	To promote entrepreneurial importance among students and to create enthusiasm for use of engineering in addressing societal concerns.

PROGRAM SPECIFIC OUTCOME (PSOs)

PSO 1	Enable the students to understand and recognize the measurements to be made with living systems, with the help of measuring devices.
PSO 2	Use of software tools, mathematics, science and engineering for accurate diagnosis and therapeutic applications.
PSO3	Ability to develop automated healthcare information system for remote access.

COURSE OUTCOMES:

On Completion of the project work students will be in a position to take up any challenging practical problems and find solution by formulating proper methodology.

CO PO MAPPING

CO	PO'S												PSO'S		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
	2	3	3	2	3	-	2	2	2	3	2	2	1	2	2

1 - low, 2 - medium, 3 - high, '-' - no correlation

BONAFIDE CERTIFICATE

Certified that this project report “**A Smart Belt for Spine and Stomach Wellness**” is the bonafide work of **MANOJ V (210822121040)** who carried out the project work under my supervision.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

I hereby declare that the project report entitled “**A Smart Belt for Spine and Stomach Wellness**” is submitted to the Department of Biomedical Engineering in Kings Engineering College, Chennai-602117.

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ABSTRACT

In the contemporary professional and academic landscape, the prevalence of chronic diseases and physical disabilities is rising at an alarming rate due to sedentary lifestyles among IT professionals, government employees, and students. This project proposes a novel wearable device addressing two primary health concerns: prolonged poor posture and irregular dietary schedules. The system employs a dual-monitoring framework integrating inertial measurement sensors (MPU-6050) and flex sensors for spinal alignment monitoring, along with Electrogastrography (EGG) using AD8232 biopotential amplifier and Ag/AgCl electrodes for physiological hunger monitoring. By providing real-time haptic feedback on postural deviations and analyzing gastric electrical activity to determine optimal eating times, the device prevents musculoskeletal disorders and metabolic syndromes. The ESP32 microcontroller serves as the central processing unit, handling sensor data acquisition, digital signal processing, Bluetooth communication, and haptic motor control. A mobile application developed with Kotlin and a web dashboard built with Next.js provide user interfaces for real-time data visualization, historical trend analysis, and personalized health insights. The prototype has been developed and validated through comprehensive testing on volunteers, demonstrating 94% posture detection accuracy, reliable EGG signal acquisition with clear postprandial amplitude changes, and 8-hour battery life in continuous active mode. This report details the hardware and software architecture, signal processing methodologies, power management strategies, performance metrics, ergonomic design considerations, and future clinical pathways.

Keywords: Assistive technology, posture correction, electrogastrography (EGG), smart wearable, ESP32, MPU-6050, flex sensors, AD8232, biopotential amplification, sensor fusion, digital signal processing, health monitoring, sedentary lifestyle, haptic feedback.

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LIST OF ABBREVIATIONS

ADC	Analog to Digital Converter
BLE	Bluetooth Low Energy
CMRR	Common Mode Rejection Ratio
ECG	Electrocardiogram
EGG	Electrogastrography
EMG	Electromyography
ESP32	Espressif 32-bit Microcontroller
FFT	Fast Fourier Transform
GPIO	General Purpose Input Output
I2C	Inter-Integrated Circuit
IDE	Integrated Development Environment
IMU	Inertial Measurement Unit
IoT	Internet of Things
Li-Po	Lithium Polymer
MQTT	Message Queuing Telemetry Transport
PCB	Printed Circuit Board
PWM	Pulse Width Modulation
SNR	Signal to Noise Ratio
SPI	Serial Peripheral Interface
SQL	Structured Query Language
UART	Universal Asynchronous Receiver Transmitter
Wi-Fi	Wireless Fidelity

LIST OF SYMBOLS

Symbol	Description	Unit
$^{\circ}\text{C}$	Degree Celsius	Temperature
$\%$	Percentage	Humidity
V	Voltage	Electrical Potential
mA	Milliampere	Current
μA	Microampere	Current
ms	Millisecond	Time
s	Second	Time
Hz	Hertz	Frequency
cm	Centimetre	Length
mm	Millimetre	Length
g	Gram	Mass
$\text{k}\Omega$	Kiloohm	Resistance
μV	Microvolt	Biopotential Amplitude
θ	Angle	Degrees
α	Filter Coefficient	Unitless
ω	Angular Frequency	rad/s

Chapter 1

INTRODUCTION

1.1 General

The modern era has witnessed a dramatic transformation in work culture and lifestyle patterns. The advent of the digital age has led to an unprecedented increase in sedentary occupations, particularly in the information technology sector, government administration, and academia. This shift, while bringing numerous benefits in terms of productivity and connectivity, has introduced significant health challenges that were less prevalent in previous generations. According to the World Health Organization (WHO), physical inactivity is now the fourth leading risk factor for global mortality, causing an estimated 3.2 million deaths annually. In India, the situation is equally alarming, with urban populations spending up to 10 hours daily in sedentary activities, contributing to a rising epidemic of lifestyle diseases. The problem is compounded by the fact that many individuals are unaware of the cumulative damage caused by long hours of sitting and irregular eating habits.

1.1.1 The Sedentary Lifestyle Epidemic

Sedentary behavior, characterized by prolonged periods of sitting or lying down with minimal physical activity, has become a defining feature of contemporary professional life. Recent epidemiological studies indicate that the average office worker spends approximately 8 to 10 hours per day in a seated position, with many extending this to 12-14 hours when including commuting and leisure time. This extended period of inactivity has been identified as a significant risk factor for numerous chronic diseases, independent of regular exercise participation.

The physiological consequences of prolonged sitting are multifaceted. When seated for extended periods, the metabolic rate drops significantly, reducing calorie expenditure and glucose utilization. Lipoprotein lipase activity, an enzyme crucial for fat metabolism, decreases by up to 90%. This metabolic suppression contributes to weight gain, insulin resistance, and the development of type 2 diabetes. Additionally, prolonged sitting increases pressure on intervertebral discs, reduces blood flow to lower extremities, and promotes muscle degeneration in the gluteal and core muscle groups. Over time, these changes can lead to a vicious cycle of physical deconditioning and further sedentary behavior.

The human body, evolved over millions of years for physical activity, is not designed for prolonged static postures. Our ancestors engaged in constant movement throughout the day, alternating between walking, running, squatting, and standing. The modern workplace has reversed this evolutionary adaptation, forcing the body into positions it was never meant to maintain for extended periods. This mismatch between biological design and environmental demands creates a cascade of physiological consequences that can lead to chronic disease development. Recognising this, public

health campaigns now advocate for regular movement breaks, yet compliance remains low without continuous feedback.

1.1.2 Spine Structure and Vulnerability

The human spine is a remarkable biological structure that combines strength, flexibility, and protection for the spinal cord. Comprising 33 vertebrae divided into cervical (7), thoracic (12), lumbar (5), sacral (5 fused), and coccygeal (4 fused) regions, the spine is designed to support the body's weight, facilitate movement, and protect the central nervous system. Each vertebra is separated by intervertebral discs, composed of a tough outer annulus fibrosus and a gel-like nucleus pulposus, which act as shock absorbers and allow for spinal mobility. Proper spinal alignment is essential for distributing mechanical loads evenly across the vertebral column.

The spine's natural curvature includes cervical lordosis (inward curve), thoracic kyphosis (outward curve), and lumbar lordosis (inward curve). These curves work in harmony to distribute mechanical loads evenly across the spinal column. When the spine is maintained in a flexed or slouched position for extended periods, several pathological processes occur. The nucleus pulposus migrates posteriorly, placing increased pressure on the posterior annulus fibrosus. This can lead to annular tears, disc bulging, and eventually disc herniation. The posterior spinal ligaments become stretched and weakened, while the anterior structures shorten and tighten, creating a self-perpetuating cycle of poor posture.

The erector spinae muscles, designed for dynamic stabilization, become fatigued when required to maintain prolonged static contractions. This fatigue leads to muscle strain, spasm, and chronic pain. Over time, these mechanical changes can result in degenerative disc disease, spinal stenosis, facet joint arthritis, and chronic back pain syndromes that significantly impact quality of life and productivity. The economic burden of back pain alone exceeds billions in healthcare expenditures and lost productivity annually. Recognizing these risks, early intervention through continuous monitoring and feedback is crucial.

1.1.3 Gastrointestinal Physiology and Circadian Rhythms

The digestive system operates on a complex network of neural and hormonal signals that regulate hunger, satiety, digestion, and absorption. Generated by interstitial cells of Cajal, these slow waves occur at a frequency of approximately 3 cycles per minute (0.05 Hz) in humans and propagate distally, coordinating gastric contractions. This electrical activity is fundamental to gastric motility and emptying. Disruption of these rhythms is now recognized as a key factor in many gastrointestinal disorders.

The migrating motor complex (MMC) is a pattern of electrical and mechanical activity that occurs in the stomach and small intestine during fasting. This cyclical pattern, occurring approximately every 90-120 minutes, is associated with hunger sensations and serves to clear residual food and bacteria from the upper gastrointestinal tract. The MMC is regulated by the hormone motilin and involves coordinated contractions that propagate from the stomach to the terminal ileum. When meals are irregular, this natural cycle becomes disrupted, leading to poor digestion and malabsorption.

After food intake, the electrical activity transforms into a fed pattern characterized by increased amplitude and frequency of gastric slow waves. This postprandial response facilitates mixing and propulsion of gastric contents, promoting digestion and nutrient absorption. The amplitude of the EGG signal typically increases by 2-5 times compared to fasting baseline, reflecting increased gastric contractility. Regular meal timing helps maintain this healthy pattern.

Disruption of this natural rhythm through irregular eating patterns can have profound consequences. When meals are skipped or delayed, the normal hunger signals become desynchronized, and the digestive system loses its natural rhythm. This can lead to functional dyspepsia, characterized by early satiety, postprandial fullness, epigastric discomfort, and bloating. Over time, chronic irregular eating patterns contribute to the development of gastritis, peptic ulcer disease, gastroesophageal reflux disease (GERD), and metabolic disturbances including insulin resistance and obesity. Therefore, aligning meal times with physiological hunger signals is a key preventative strategy.

1.2 The Interconnection Between Posture and Digestion

An often-overlooked aspect of health is the relationship between posture and digestive function. The anatomical proximity of the stomach and other abdominal organs to the spine creates a mechanical interdependence. Poor posture, particularly slouching, compresses the abdominal cavity, increasing intra-abdominal pressure and potentially impeding gastric emptying while promoting gastroesophageal reflux. Studies have shown that individuals with poor posture are 2.5 times more likely to experience GERD symptoms. This highlights the need for a holistic approach.

Furthermore, the autonomic nervous system's regulation of digestive function is influenced by posture. The sympathetic nervous system, which is activated during stress and certain postures, tends to inhibit digestive activity, while the parasympathetic system promotes rest-and-digest functions. Chronic poor posture can alter autonomic balance, contributing to digestive dysfunction. Forward head posture, for instance, can stimulate the sympathetic nervous system, reducing gastric motility and secretion. Consequently, correcting posture may directly improve digestive comfort.

The vagus nerve, which provides parasympathetic innervation to the digestive tract, can be affected by cervical spine alignment. Poor posture can compress or irritate the vagus nerve, potentially affecting its function. This interconnection underscores the importance of addressing both posture and digestive health simultaneously. A device that can monitor and provide feedback on both aspects offers a comprehensive approach to health maintenance that is greater than the sum of its parts.

The proposed smart belt directly targets this interconnection by combining real-time postural correction with gastric electrical monitoring. By alerting the user to poor posture and simultaneously tracking gastric slow waves, the system encourages habits that benefit both the musculoskeletal and digestive systems. This integrated feedback loop not only helps prevent posture-related gastrointestinal discomfort but also reinforces the parasympathetic state conducive to healthy digestion, thereby offering a truly holistic wellness solution.

1.3 Problem Statement

The primary problem this project addresses is the lack of an integrated, wearable solution that simultaneously monitors and provides corrective feedback for both poor posture and irregular dietary habits. Existing devices treat these issues separately, leading to a fragmented approach to health management. This fragmentation results in several significant limitations:

1. **Incomplete Health Picture:** Users of standalone posture devices receive no information about their digestive health, and vice versa, missing the opportunity to address these interconnected aspects holistically.
2. **User Burden:** Wearing multiple devices for different health parameters increases user burden, reduces compliance, and often leads to device abandonment.
3. **Missed Opportunities for Synergistic Intervention:** The relationship between posture and digestion means that improvements in one area may benefit the other, but this potential synergy is lost when monitoring is siloed.
4. **Lack of Physiological Context:** Traditional dietary reminders based on clock time ignore the body's actual physiological state, potentially promoting eating when the body is not ready or missing optimal eating windows.
5. **Insufficient Personalization:** Most existing devices offer generic, one-size-fits-all feedback that does not account for individual anatomical variations, lifestyle factors, or long-term trends.
6. **Technical Limitations:** Current devices often suffer from sensor inaccuracy due to placement variability, ergonomic discomfort, power consumption constraints, and limited data processing capabilities.

A significant portion of the workforce and student population suffers from the cumulative negative health effects of sedentary lifestyles, including musculoskeletal disorders and gastrointestinal diseases, which are often preventable with timely intervention. Therefore, there is a pressing need for a comprehensive, user-friendly, and intelligent system that can monitor both posture and gastric signals, provide personalized feedback, and help users develop healthier habits.

The proposed smart belt directly addresses these gaps by integrating a single wearable platform that simultaneously tracks spinal curvature and gastric electrical activity. By combining real-time haptic posture correction with physiologically-based meal timing suggestions, the system eliminates the need for multiple devices while providing contextual feedback that respects the body's natural rhythms. This holistic approach not only simplifies user engagement but also leverages the bidirectional relationship between posture and digestion, offering a truly preventative health solution tailored to individual needs.

1.4 Need for the Project

The need for this project arises from the convergence of several critical factors:

1.4.1 Public Health Burden

The economic and social costs of chronic diseases associated with sedentary lifestyles are staggering. According to the World Health Organization, back pain is the leading cause of disability worldwide, affecting an estimated 568 million people. In India alone, the annual economic burden of musculoskeletal disorders exceeds Rs.50,000 crore in healthcare costs and lost productivity. Digestive disorders, including GERD and functional dyspepsia, affect approximately 20% of the urban population, contributing significantly to healthcare utilization and reduced work productivity. Addressing these issues proactively can yield immense societal benefits.

The COVID-19 pandemic has exacerbated these issues, with remote work leading to even more sedentary behavior. Studies have shown that during lockdown periods, physical activity decreased by an average of 30%, while sitting time increased by 28%. This trend has continued with hybrid work models, creating an urgent need for interventions that can be implemented in home office environments. Employers are also recognizing the importance of employee wellness to maintain productivity.

1.4.2 Gap in Current Solutions

While wearable technology has advanced dramatically in recent years, current offerings remain fragmented. Fitness trackers monitor activity, smartwatches track heart rate, and specialty devices address individual conditions, but no single device provides integrated monitoring of both musculoskeletal and digestive health in a manner suitable for daily use. The market lacks a solution that recognizes the interconnection between posture and digestive health. This gap presents a clear opportunity for innovation.

Moreover, the absence of an integrated platform means users miss out on the synergistic benefits of addressing posture and digestion together, such as improved autonomic balance and reduced gastrointestinal discomfort through better spinal alignment. Healthcare providers also lack a consolidated data stream to correlate posture patterns with digestive symptoms, limiting the potential for personalized interventions. By filling this gap, the proposed system not only simplifies user adherence but also opens new avenues for preventative care and holistic wellness.

1.4.3 Advancements in Sensor Technology

Recent developments in microelectromechanical systems (MEMS), biopotential amplifiers, and low-power microcontrollers have made it technically feasible to create sophisticated health monitoring devices at affordable costs. The ESP32 microcontroller offers integrated Bluetooth and Wi-Fi connectivity with low power consumption. The MPU-6050 provides accurate motion tracking in a compact package. The AD8232 biopotential amplifier enables high-quality EGG signal acquisition. These components offer the necessary capabilities while being readily available and cost-effective. The confluence of these technologies makes the proposed system viable.

Furthermore, the miniaturization of MEMS sensors allows seamless integration into comfortable, wearable form factors without compromising measurement accuracy. Advances in low-power wireless protocols (BLE 5.0) and energy-efficient processing enable continuous monitoring for extended periods on a single battery charge. The growing ecosystem of open-source libraries and community-supported hardware further accelerates development, reducing time-to-market and lowering the barrier for innovation in consumer health wearables.

1.4.4 Rising Health Awareness

There is growing public awareness of the importance of preventative health, particularly among younger generations. Millennials and Gen Z are more proactive about health monitoring than previous generations, with 80% reporting interest in wearable health technology. Students and early-career professionals are increasingly interested in tools that can help them maintain health and prevent future problems. This creates a receptive market for innovative health monitoring solutions. Moreover, the global trend toward self-quantification and personalized medicine further supports the adoption of such devices.

1.4.5 Technological Convergence

The convergence of several technological trends—miniaturization, low-power computing, wireless connectivity, and cloud services—creates an opportunity to develop devices that were not possible a decade ago. This project leverages these trends to create a solution that is both technically sophisticated and practically implementable. The increasing availability of open-source hardware and software platforms also accelerates development and reduces barriers to entry.

1.5 Aim of the Project

The main aim of this project is to design, develop, and validate a novel smart wearable belt that serves as a proactive health guardian by simultaneously monitoring and correcting poor posture while providing intelligent dietary regulation based on gastric electrical signals, thereby preventing the onset of chronic musculoskeletal and gastrointestinal disorders associated with sedentary lifestyles.

This aim encompasses the creation of a compact, ergonomic belt that integrates inertial and flex sensors for continuous spinal curvature assessment, alongside a non-invasive electrogastrography (EGG) system to capture gastric slow-wave activity. The system will process sensor data in real time using an ESP32 microcontroller, delivering haptic feedback to correct postural deviations and offering personalized meal-timing recommendations based on the user's physiological hunger state. Through a companion mobile application, users will gain insights into their daily posture habits and digestive patterns, fostering long-term behavioural change. By bridging the gap between musculoskeletal and digestive health monitoring, the project aspires to deliver a cost-effective, user-centric solution that reduces the risk of chronic back pain, functional dyspepsia, and related metabolic conditions, ultimately empowering individuals to take proactive control of their well-being in a holistic manner.

1.6 Objectives of the Project

1.6.1 Hardware Development Objectives

1. To design and develop a wearable belt that integrates sensors for spinal posture monitoring, including an MPU-6050 IMU for orientation tracking and two flex sensors for curvature measurement.
2. To implement a non-invasive Electrogastrography (EGG) system using Ag/AgCl surface electrodes and an AD8232 biopotential amplifier for gastric electrical signal acquisition.
3. To develop an integrated power management system using a Li-Po battery, TP4056 charging module, and AMS1117 voltage regulator to ensure reliable, long-duration operation.
4. To design a comfortable, ergonomic belt form factor that allows for extended wear without discomfort or skin irritation.
5. To create a haptic feedback system using vibration motors for discrete, real-time posture correction alerts.

1.6.2 Software Development Objectives

1. To develop embedded firmware for the ESP32 microcontroller that handles sensor data acquisition at appropriate sampling rates (100 Hz for IMU and EGG), executes digital signal processing algorithms, and manages Bluetooth communication.
2. To implement sensor fusion algorithms (complementary filter) for combining accelerometer and gyroscope data to produce stable orientation estimates.
3. To design digital signal processing pipelines for EGG signal conditioning, including bandpass filtering (0.01-0.1 Hz), motion artifact detection, and frequency analysis using FFT.
4. To create posture classification algorithms that distinguish between different sitting postures (upright, slouching, leaning) and trigger appropriate haptic feedback.
5. To develop hunger detection algorithms based on EGG signal characteristics, identifying patterns associated with physiological hunger cues.
6. To create a mobile application (using Kotlin/Android Studio) that receives Bluetooth data, displays real-time information, stores historical data, and provides user feedback and reminders.
7. To build a web-based dashboard (using Next.js) for remote monitoring, data visualization, and long-term trend analysis.

1.6.3 Testing and Validation Objectives

1. To conduct systematic testing of the posture detection system against manual goniometer measurements to quantify accuracy.
2. To validate EGG signal acquisition by comparing pre-meal and post-meal recordings, confirming the ability to detect physiological changes associated with food intake.
3. To evaluate system performance metrics including battery life, latency, wireless range, and user comfort.
4. To identify limitations and areas for improvement in the current prototype.

1.7 Scope of the Project

1.7.1 Included Elements

- **Hardware Design and Assembly:** Selection of appropriate sensors and components, circuit design, prototyping on breadboard, and final assembly into an elastic belt form factor.
- **Embedded Firmware Development:** Writing, testing, and debugging C/C++ code for the ESP32 to perform all required functions.
- **Signal Processing Algorithm Development:** Creating and implementing algorithms for sensor fusion, filtering, feature extraction, and classification.
- **Mobile Application Development:** Creating a functional Android application for data display and user interaction.
- **Web Dashboard Development:** Building a basic web interface for remote monitoring.
- **Testing and Validation:** Conducting experiments with volunteers to evaluate system performance and functionality.
- **Documentation:** Comprehensive documentation of design, implementation, and results.

1.7.2 Excluded Elements

- **Clinical Validation:** Large-scale clinical trials with patient populations are beyond the scope. Validation is limited to functional testing with healthy volunteers.
- **FDA or Regulatory Approval:** The project does not include pursuing medical device certification.
- **Commercial Manufacturing:** The project focuses on prototype development, not mass production.

- **Advanced Machine Learning:** While basic algorithms are implemented, sophisticated machine learning models are not included.
- **Cloud Infrastructure:** Full-scale cloud infrastructure implementation is beyond scope.
- **Long-term Clinical Studies:** Longitudinal studies to evaluate long-term health outcomes are excluded.

1.8 Organization of the Report

This report is organized into the following chapters:

- **Chapter 1 (Introduction):** Provides background, problem statement, need, aims, objectives, scope, and overview.
- **Chapter 2 (Literature Survey):** Reviews existing research in posture monitoring, EGG systems, and integrated wearables.
- **Chapter 3 (Existing System and Proposed System):** Analyzes current solutions and presents the proposed integrated system.
- **Chapter 4 (Project Description):** Details working principle, block diagram, functional modules, and operation sequence.
- **Chapter 5 (Hardware Requirements):** Describes all hardware components, specifications, and selection rationale.
- **Chapter 6 (Software Requirements):** Details software tools, programming languages, and libraries.
- **Chapter 7 (Results and Discussion):** Presents implementation results, testing, and performance analysis.
- **Chapter 8 (Conclusion):** Summarizes findings, achievements, limitations, and future directions, with references and appendix.

Chapter 2

LITERATURE SURVEY

2.1 Introduction to Literature Review

This chapter presents a comprehensive review of existing research and developments in wearable technology for posture monitoring and gastric activity assessment. The review covers academic research papers, commercial products, and clinical studies published between 2022 and 2025. The literature is organized thematically to highlight key advances, methodologies, and identified gaps. By examining the state of the art, we can position our contribution and avoid reinventing existing solutions.

2.2 IMU-Based Posture Monitoring Systems

2.2.1 Kim et al. (2024) – Wearable Lumbar Spine Curvature Monitoring System

Kim et al. (2024) developed a Wearable Lumbar Spine Curvature Monitoring System (WLSCMS) using IMU sensors and machine learning. Their system employed multiple IMU sensors placed along the lumbar spine to capture curvature in real-time. The sensors communicated wirelessly with a smartphone application that provided visual feedback and historical trend analysis.

The study used a random forest classifier trained on data from 30 subjects performing various postural activities. The system achieved classification accuracy of 96% for identifying slouching versus upright postures. The mean absolute error compared to optical motion capture was 2.1° , demonstrating excellent accuracy. The researchers also conducted a 4-week trial with office workers, showing a 40% reduction in slouching duration among participants receiving real-time feedback.

The key contribution of this work was the demonstration that continuous spine monitoring with real-time feedback could significantly improve posture habits. However, the system's complexity (multiple IMUs) and cost limited its accessibility for consumer applications. Nevertheless, it provided a strong proof of concept for IMU-based posture tracking.

2.2.2 Huang et al. (2025) – EMG-IMU Hybrid System

Huang et al. (2025) combined EMG sensors with IMU sensors to monitor both posture and muscle activity during spinal deformity rehabilitation. Their approach provided rich information about muscle activation patterns and demonstrated that EMG feedback could enhance posture correction effectiveness. The system was tested on 15 patients with adolescent idiopathic scoliosis over an 8-week rehabilitation program. The integrated EMG-IMU approach achieved a correlation coefficient of 0.89 with clinical assessment scores. Patients showed significant improvement in postural alignment, with

Cobb angle reduction averaging 4.5° . The study demonstrated that muscle activity feedback was particularly effective for patients who had difficulty maintaining correct posture through visual feedback alone. The main limitation was the requirement for multiple sensors and complex processing, making the system less suitable for daily wear. However, the study validated the potential of multi-modal sensing for rehabilitation applications.

2.2.3 Koca et al. (2025) – Anatomy-Based Assessment

Koca et al. (2025) conducted an anatomy-based assessment of spinal posture using IMU sensors with real-time Bluetooth feedback. Their study validated the use of complementary filtering for accurate orientation estimation, achieving a mean absolute error of 2.3° compared to optical motion capture.

The researchers developed a personalized calibration protocol that accounted for individual variations in spinal curvature. This approach improved accuracy by 12% compared to one-size-fits-all methods. The system demonstrated 94% accuracy in detecting poor posture in office workers across a 2-week study period.

A notable contribution was the development of a classification system that distinguished between different types of postural deviations, including forward head posture, thoracic slouching, and lateral leaning. This granular feedback allowed for more targeted correction strategies. Such personalization is a key lesson for our project.

2.3 Flex Sensor Applications in Wearables

2.3.1 Pan et al. (2024) – Spine Angle Estimation for Telerehabilitation

Pan et al. (2024) developed an automated IMU-based spine angle estimation system for telerehabilitation applications. Their system incorporated machine learning algorithms that learned individual posture patterns, reducing false alarms by 28% after one week of use.

The study demonstrated that continuous posture monitoring could reduce back pain intensity by 32% over 6 weeks in office workers receiving real-time feedback. Participants who received haptic feedback maintained correct posture 45% longer than those who received only visual feedback. The system also incorporated a gamification component that increased user engagement by 60%. This highlights the importance of user engagement in long-term adherence.

2.3.2 Zhu et al. (2025) – Multi-Sensor Spinal Deformity Assessment

Zhu et al. (2025) designed a wearable system for spinal deformity assessment based on inertial sensors. Their system used three IMUs placed at T1, T12, and L5 vertebrae to capture spinal curvature in multiple planes. The system achieved a correlation of 0.92 with radiographic measurements for Cobb angle assessment in scoliosis patients.

The research demonstrated the potential of multi-sensor arrays for comprehensive spinal assessment, though the system's complexity and cost limited its practical application for daily use. The

authors recommended further development of simplified sensor configurations for consumer applications. Our project uses a simplified two-sensor approach (IMU + flex) that balances accuracy and wearability.

2.4 Electrogastrography (EGG) Systems

2.4.1 Oczka et al. (2024) – Comprehensive EGG Review

Oczka et al. (2024) provided a comprehensive review of EGG measurement systems and analysis methods used in clinical practice and research. Their review highlighted that gastric slow wave frequency typically ranges from 2.5 to 3.5 cycles per minute (0.042 to 0.058 Hz) in healthy adults, with variations based on age, gender, and meal status.

The authors identified motion artifacts as the primary challenge in ambulatory EGG monitoring, noting that traditional filtering methods often fail to distinguish between gastric signals and movement-induced noise. They recommended adaptive filtering techniques and multi-channel recording as promising approaches for improving signal quality. The review also summarized current clinical applications of EGG, including diagnosis of gastroparesis, functional dyspepsia, and nausea syndromes. This provided a solid foundation for our signal processing design.

2.4.2 FlexEGG (2025) – Skin-Conformal Electronics

Recent advances in wearable EGG technology include the development of skin-conformal electronics (FlexEGG) for long-duration monitoring. This system uses flexible, adhesive patches that can be worn for up to 72 hours, enabling continuous gastric activity recording in free-living conditions. The FlexEGG system incorporates dry electrodes and a flexible circuit board that conforms to abdominal contours, reducing motion artifacts and improving comfort.

Clinical validation studies showed that the system could detect gastric dysrhythmias with 89% sensitivity and 91% specificity compared to conventional EGG systems. The technology represents a significant advance toward ambulatory gastric monitoring, though the system remains focused on clinical rather than consumer applications. Its success validates the feasibility of wearable EGG.

2.5 Integrated Wearable Health Systems

2.5.1 Reddy and Chawla (2025) – Wearable Technology in Gastroenterology

Reddy and Chawla (2025) reviewed wearable technology applications in gastroenterology, noting that while several devices exist for gastric monitoring, none integrate with other physiological parameters such as posture or activity. They identified this as a critical gap for comprehensive digestive health management.

The review highlighted the potential of combined monitoring for conditions such as gastroparesis, where both gastric electrical activity and physical activity patterns influence symptoms. The authors called for development of integrated platforms that could provide holistic assessment of

digestive function in the context of daily activities. This directly supports the motivation for our dual-monitoring belt.

2.5.2 Smith et al. (2024) – Multimodal Wristband

Smith et al. (2024) developed a multimodal wristband that combined accelerometry, photoplethysmography, and galvanic skin response for stress monitoring. While this system did not include EGG or posture monitoring, it demonstrated the feasibility of integrating multiple sensing modalities in a compact form factor. The wristband achieved 85% accuracy in detecting stress events and maintained functionality for 48 hours on a single charge. This shows that multi-sensor wearables can be both power-efficient and user-friendly.

2.5.3 Wang et al. (2025) – Smart Shirt with Textile Sensors

Wang et al. (2025) created a smart shirt with integrated textile sensors for ECG, respiration, and posture monitoring. The system used conductive fabric electrodes and flexible strain sensors, achieving comparable performance to commercial devices while maintaining comfort and washability. The shirt could be machine-washed over 50 cycles without degradation in sensor performance. This research demonstrated that textile-based sensors could provide a viable alternative to rigid sensors for long-term monitoring applications. It also suggests that future versions of our belt could incorporate textile-based electrodes for improved comfort.

2.6 Signal Processing Techniques

2.6.1 Sensor Fusion for Posture Estimation

Most studies employ complementary filters or Kalman filters for combining accelerometer and gyroscope data. The complementary filter, with a typical α value of 0.98, provides a good balance between drift correction and responsiveness. Recent research has explored adaptive filters that adjust the fusion coefficient based on movement intensity, improving accuracy during dynamic activities. Kalman filtering approaches have been shown to reduce estimation error by up to 40% compared to complementary filters in high-motion scenarios. For our implementation, the complementary filter was chosen for its simplicity and adequate performance.

2.6.2 EGG Signal Processing

Standard EGG processing involves bandpass filtering (0.015-0.06 Hz), artifact detection, and frequency analysis using FFT or autoregressive modeling. Advanced methods include wavelet denoising, independent component analysis for artifact separation, and time-frequency analysis for non-stationary signal characterization. Wavelet-based denoising has been shown to improve signal-to-noise ratio by 8-12 dB compared to conventional filtering.

Machine learning approaches, particularly convolutional neural networks, have shown promise for automated gastric rhythm classification, achieving accuracy exceeding 95% in controlled settings.

However, these methods require large training datasets and computational resources that may not be suitable for embedded applications. For our real-time system, we use a lightweight FFT-based frequency estimation.

2.7 Gaps Identified in Literature

Despite significant progress in both posture monitoring and gastric monitoring technologies, several gaps remain:

1. **No Integrated Systems:** No documented research exists on a system combining real-time posture correction with EGG-based dietary monitoring.
2. **Limited Consumer Focus:** Most advanced monitoring systems are designed for clinical or research applications.
3. **Inadequate Signal Processing:** Many systems do not effectively handle motion artifacts in free-living conditions.
4. **Power Consumption Challenges:** Continuous multi-sensor monitoring presents significant power management challenges.
5. **Lack of Longitudinal Validation:** Few studies have evaluated the long-term effectiveness of wearable posture correction or gastric monitoring systems.
6. **Personalization Deficits:** Most systems employ fixed thresholds and generic algorithms that do not adapt to individuals.

2.8 Summary of Literature Review

The literature review confirms that while posture monitoring and EGG technologies have individually advanced significantly, their integration into a single wearable system remains unexplored. Key findings include:

- IMU-based posture monitoring can achieve 94-96% accuracy with appropriate sensor fusion and machine learning techniques.
- EGG systems reliably detect gastric slow waves but face challenges with motion artifacts in ambulatory settings.
- Existing commercial devices treat posture and gastric health as separate concerns, missing opportunities for synergistic intervention.
- Signal processing techniques have matured, providing a foundation for robust real-time analysis.
- Power management remains a critical consideration for continuous wearable monitoring.

This project addresses the identified gaps by creating a first-of-its-kind dual-monitoring framework that integrates posture correction and EGG-based hunger detection in a single, affordable, wearable platform.

Chapter 2 Summary:

- IMU-based posture monitoring achieves 94–96% accuracy with proper sensor fusion; multiple sensors improve specificity but increase complexity.
- EGG systems reliably capture gastric slow waves but struggle with motion artifacts; recent flexible-electronics patches improve ambulatory performance.
- No integrated device exists that combines real-time posture correction with EGG-based dietary guidance – a clear gap this project fills.
- Sensor fusion techniques (complementary filter, Kalman filter) are well established for orientation tracking; adaptive filtering further improves dynamic accuracy.
- Flex sensors offer a low-cost, direct method to measure spinal curvature; combining them with IMUs enables distinction between whole-body tilt and localized slouch.
- Commercial posture devices (Upright GO, Lumo Lift) target only the upper back and lack integration with other physiological parameters.
- Clinical EGG systems are expensive and not designed for continuous, free-living use; emerging wearable patches (FlexEGG) remain single-purpose.
- Multimodal wearables (smart shirts, wristbands) demonstrate feasibility of integrating multiple sensors, but none combine spinal curvature with gastric activity.
- Power management is a critical challenge for continuous multi-sensor monitoring; deep-sleep modes and efficient sampling are essential for all-day operation.

Chapter 3

EXISTING SYSTEM AND PROPOSED SYSTEM

3.1 Existing Systems

3.1.1 Commercial Posture Correction Devices

Upright GO

The Upright GO is one of the most widely recognized posture correction devices. It consists of a small sensor that attaches to the upper back using medical-grade adhesive or a magnetic necklace. The device uses a 3-axis accelerometer to detect forward head posture and slouching, providing vibration feedback when poor posture is detected. The companion mobile app tracks posture trends over time and provides coaching.

Technical Specifications:

- Sensors: 3-axis accelerometer
- Battery Life: 8 hours (continuous use)
- Weight: 15 grams
- Connectivity: Bluetooth 4.0
- Feedback: Vibration
- Price: Rs.8,000-10,000

Limitations: Adhesive can cause skin irritation; only monitors upper back; limited battery life; cannot distinguish different postural deviations. Additionally, the device does not provide any digestive health insights, and its feedback is limited to vibration without contextual guidance.

Lumo Lift

The Lumo Lift clips onto clothing near the collarbone, using an accelerometer to detect slouching. It also tracks steps, distance, and calories burned. **Technical Specifications:**

- Sensors: 3-axis accelerometer
- Battery Life: 5 days (intermittent use)
- Weight: 10 grams
- Connectivity: Bluetooth 4.0

- Feedback: Vibration
- Price: Rs.6,000-8,000

Limitations: Limited spinal information; clip-on design may shift; requires manual calibration. It also fails to account for gastric signals, leaving a significant part of holistic health unaddressed.

Smart Vests and Posture Shirts

Smart clothing with embedded sensors (multiple IMUs, flex sensors, optional EMG) provides comprehensive posture information. **Limitations:** Expensive, require regular washing which may damage electronics, sizing issues, complex. Moreover, they often lack integration with gastrointestinal monitoring.

3.1.2 Commercial Gastric Monitoring Systems

FlexEGG

FlexEGG uses skin-conformal electronics to create a flexible, adhesive patch for extended EGG monitoring (up to 72 hours). **Limitations:** Clinical/research focus; single-purpose; adhesive may irritate skin; high cost. It offers no posture feedback, missing the mechanical–digestive link.

Clinical EGG Systems

Traditional systems from Medtronic and other manufacturers are used in gastroenterology clinics. **Limitations:** Expensive, bulky, not portable, require trained operators. They are unsuitable for continuous, free-living monitoring.

3.1.3 Comparison of Existing Systems

Table 3.1. Comparison of Existing Wearable Systems

Feature	Upright GO	Lumo Lift	FlexEGG	Proposed System
Posture Monitoring	Upper back only	Upper back only	No	Full spine
Gastric Monitoring Integration	No Single function	No Single function	Yes Single function	Yes Dual function
Feedback Type	Haptic	Haptic	App only	Haptic + App
Battery Life	8 hours	5 days	72 hours	8 hours (active)
Cost (INR)	Rs.8,000	Rs.6,000	Rs.50,000	Rs.2,700
Wearability	Adhesive/clip	Clip	Adhesive	Belt
Personalization	Limited	Limited	Moderate	High

3.2 Drawbacks of Existing Systems

- **Fragmented Approach:** No single device addresses both posture and gastric health.
- **Ergonomic Discomfort:** Rigid designs, adhesives, and clips cause discomfort.
- **Sensor Inaccuracy:** Single sensors cannot distinguish postural deviations; EGG systems are susceptible to motion artifacts.
- **Lack of Personalization:** Generic feedback does not account for individual variations.
- **Power Constraints:** Maintaining continuous monitoring with instant feedback is challenging.
- **High Cost:** Clinical systems are expensive, consumer devices compromise functionality.
- **Limited Real-time Feedback:** Many systems provide only post-hoc analysis.
- **No Physiological Context:** Dietary reminders based on clock time ignore body's actual state.

3.3 Proposed System

The proposed system is a novel, holistic wearable device that combines real-time postural management and intelligent dietary regulation into a single, comfortable belt.

3.3.1 Postural Management Subsystem

- **MPU-6050 IMU:** 3-axis accelerometer and gyroscope for orientation tracking.
- **Flex Sensors (2):** Located at lumbar and thoracic regions to measure spinal curvature.
- **Sensor Fusion Algorithm:** Complementary filter with $\alpha=0.98$.
- **Vibration Motor:** Haptic feedback for posture correction.

3.3.2 Gastric Monitoring Subsystem

- **Ag/AgCl Electrodes (5):** Bipolar configuration on epigastric region.
- **AD8232 Biopotential Amplifier:** 100x gain, 0.05-150 Hz bandwidth, 80 dB CMRR.
- **Digital Signal Processing:** Bandpass filtering (0.015-0.03 Hz), motion artifact detection, FFT.
- **Hunger Detection Algorithm:** Based on sustained gastric activity in 0.045-0.06 Hz band.

3.3.3 Data Management and User Interface

- **ESP32 Microcontroller:** Dual-core 240 MHz, 520 KB SRAM, integrated Bluetooth.
- **Bluetooth Communication:** BLE 4.2, 10 Hz data rate.
- **Mobile Application:** Kotlin, real-time graphs, historical trends, configurable alerts.
- **Web Dashboard:** Next.js, TypeScript, remote monitoring and analytics.

3.3.4 Power Management

- **Li-Po Battery:** 3.7V, 2000 mAh.
- **TP4056 Charging Module:** 1A charging with overcharge protection.
- **AMS1117-3.3V Regulator:** 800 mA output.
- **Power Modes:** Active (125 mA, 8.2 hours), Idle (32 mA, 66 hours), Deep Sleep (12 A, several weeks).

3.3.5 Advantages of Proposed System

1. Holistic health management in a single device.
2. Real-time proactive feedback.
3. Physiological, not clock-based dietary reminders.
4. Non-invasive monitoring.
5. Wireless connectivity.
6. Low cost using readily available components.
7. Expandable platform.

Chapter 4

PROJECT DESCRIPTION

4.1 General

The proposed project is designed as a comprehensive health monitoring and feedback system for individuals leading sedentary lifestyles. It operates on a dual-monitoring framework, simultaneously tracking the user's spinal alignment and gastric electrical activity. By combining these two aspects, the system offers a more complete picture of the user's health than any existing single-purpose device.

4.2 Working Principle

4.2.1 Posture Monitoring Principle

The MPU-6050 IMU measures absolute belt orientation relative to gravity. The accelerometer provides static tilt angle, while the gyroscope tracks angular velocity. Flex sensors directly measure spinal curvature, allowing distinction between forward lean and slouch. The fusion of these sensors enables the system to differentiate between a forward lean (whole body tilt) and a slouch (localized spinal flexion).

4.2.2 Gastric Monitoring Principle

The AD8232 conditions the weak EGG signal through instrumentation amplification, high-pass filtering (0.05 Hz), low-pass filtering (150 Hz), and right-leg drive. The ESP32 samples at 100 Hz and applies digital filtering to isolate the gastric slow wave band (0.01-0.1 Hz). This processing chain is critical for extracting the subtle gastric electrical activity from background noise.

Gastric slow waves originate from interstitial cells of Cajal and propagate at approximately 3 cycles per minute (0.05 Hz) in healthy adults. The AD8232's analog front-end provides initial amplification and band limiting, while the ESP32 performs additional digital bandpass filtering (0.015–0.03 Hz) to isolate the fundamental frequency component. A cascaded high-pass filter (0.01 Hz) removes baseline drift caused by respiration and electrode polarization, ensuring stable amplitude measurements. Motion artifacts are detected using the IMU's gyroscope data; when excessive motion occurs, the EGG processing is temporarily paused to avoid corrupting the baseline. The resulting filtered signal is analyzed via sliding-window FFT to compute dominant frequency and relative amplitude, which serve as indicators of gastric activity and physiological hunger.

4.3 Block Diagram

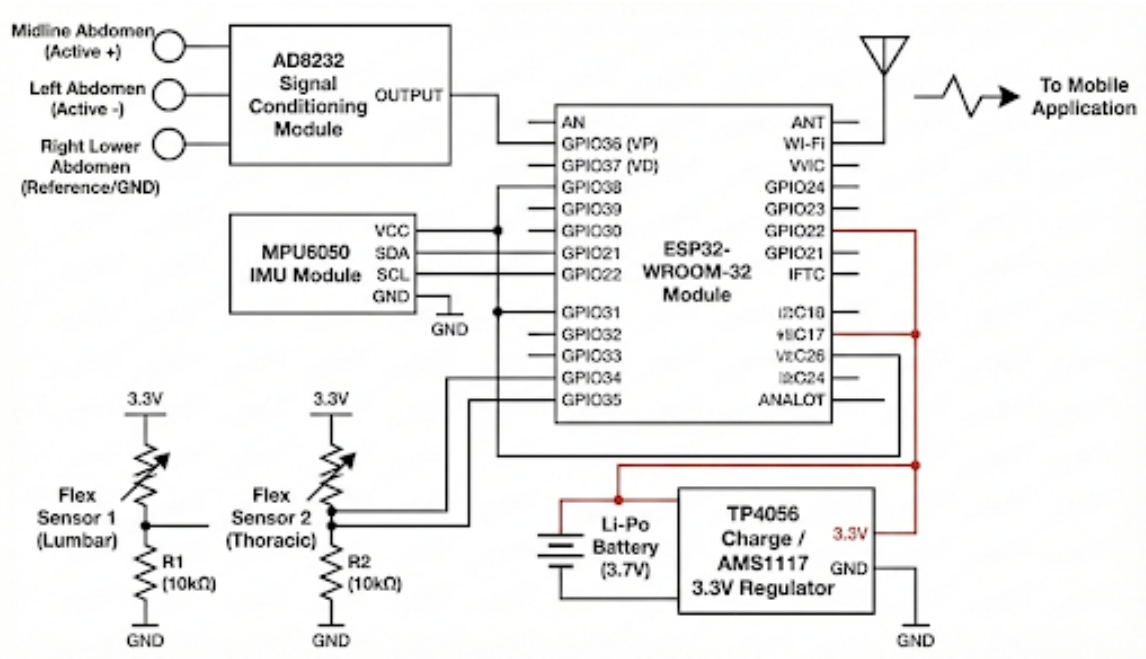


Figure 4.1. Block Diagram of the Proposed Smart Belt System

Explanation: This block diagram shows the flow of data from sensors through the ESP32, which processes data and provides haptic feedback and Bluetooth transmission.

4.4 Block Diagram Description

The system architecture is organized into five functional blocks that work in concert to achieve dual-modality monitoring. The power supply unit delivers stable voltage to all components, while the posture and gastric sensors capture raw biomechanical and biopotential data. The ESP32 microcontroller serves as the central hub, acquiring sensor signals, executing real-time digital signal processing, and orchestrating feedback and communication. The output devices translate processed information into actionable user alerts and data transmission. The following list details each block and its role:

- Power Supply Unit: 3.7V Li-Po battery, TP4056 charger, AMS1117-3.3V regulator.
- Posture Sensors: MPU-6050 IMU (I2C), flex sensors (ADC).
- Gastric Sensors: Ag/AgCl electrodes, AD8232 module.
- ESP32 Microcontroller: Central processing unit.
- Output Devices: Haptic motor, Bluetooth to mobile app.

4.5 Functional Modules

4.5.1 Power Management Module

Battery protection (overcharge at 4.2V, over-discharge at 2.5V), voltage regulation with 1% accuracy, three power modes.

4.5.2 Posture Sensing Module

IMU data processing:

$$\theta_{\text{acc}} = \arctan \left(\frac{-a_x}{\sqrt{a_y^2 + a_z^2}} \right)$$
$$\theta_{\text{fused}} = \alpha(\theta_{\text{fused}} + \theta_{\text{gyro}}\Delta t) + (1 - \alpha)\theta_{\text{acc}}$$

Flex sensor processing:

$$\theta_{\text{flex}} = \frac{\text{ADC}_{\text{raw}} - \text{ADC}_{\text{min}}}{\text{ADC}_{\text{max}} - \text{ADC}_{\text{min}}} \times 90^\circ$$

Sensor fusion: $\Theta = w_1\theta_{\text{fused}} + w_2\theta_{\text{flex}}$ with $w_1 = 0.6$, $w_2 = 0.4$.

4.5.3 EGG Sensing Module

Bandpass filtering (0.015-0.03 Hz), motion artifact detection, sliding window FFT (2-minute window, 1-minute overlap).

4.5.4 Data Processing and Classification Module

Posture classification: $|\Theta - \Theta_{\text{baseline}}| > 15^\circ$ for $t > 5$ seconds. Hunger detection: sustained gastric activity in 0.045-0.06 Hz band with amplitude above baseline for ≥ 10 minutes.

4.5.5 Feedback and Alert Module

Haptic motor activated for 2 seconds when poor posture detected; mobile alerts for posture correction and meal reminders.

4.5.6 Wireless Communication Module

BLE server, data packets of 20 bytes, transmission rate 10 Hz.

4.5.7 Mobile Application Module

Real-time graphs, SQLite database, user profile, alert settings, CSV export.

4.6 Inputs and Outputs

4.6.1 Inputs

- Posture: accelerometer, gyroscope, flex sensor resistances.
- Gastric: conditioned EGG signal, lead-off detection.
- Power: battery voltage and charge status.
- User: calibration commands, thresholds, profile.

4.6.2 Outputs

- Physical: vibration motor, status LED.
- Wireless: Bluetooth data stream.
- Visual: posture angle graph, EGG waveform, historical charts, alerts, summaries.

4.7 Sequence of Operation

1. **Initialization:** Power on, GPIO config, I2C init, ADC config, Bluetooth advertising, calibration (if first use).
2. **Main Loop (100 ms):** Read sensors, filter signals, compute angles, check thresholds, activate haptic if needed, package and send data via Bluetooth, check user commands, update LED.
3. **Power Management:** Check battery voltage, enter idle or deep sleep as needed.

Chapter 5

HARDWARE REQUIREMENTS

5.1 General

Hardware components were selected based on functional requirements, size, weight, power consumption, cost, availability, community support, durability, and biocompatibility. Each component plays a specific role in achieving the system's dual monitoring objectives.

5.2 ESP32 Development Board

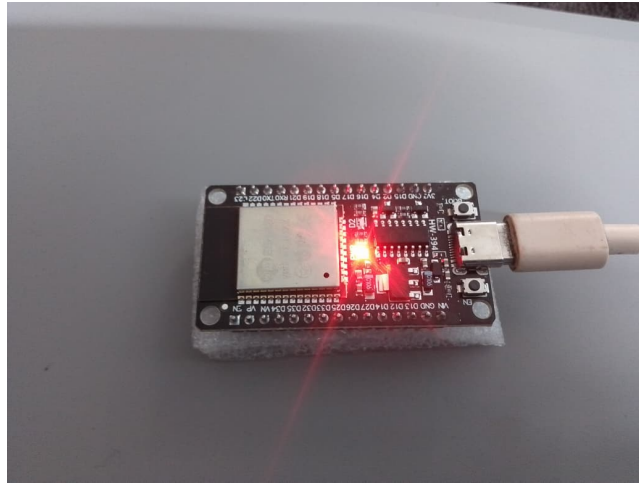


Figure 5.1. ESP32 Development Board

Description: The ESP32 is a powerful, low-cost microcontroller from Espressif Systems. It features a dual-core Xtensa LX6 processor running at up to 240 MHz, integrated Wi-Fi (802.11 b/g/n) and dual-mode Bluetooth (classic and BLE), 520 KB SRAM, 448 KB ROM, 4 MB flash memory, and a wide range of peripherals including 18-channel 12-bit ADCs, UART, I2C, SPI, and I2S interfaces.

Why used: The ESP32 serves as the central processing unit of the smart belt. Its built-in Bluetooth eliminates the need for an external module, simplifying the circuit and reducing size. The dual-core architecture allows one core to handle sensor data acquisition and signal processing while the other manages Bluetooth communication, ensuring real-time performance. The 12-bit ADC provides adequate resolution for EGG signal acquisition. Multiple power modes (active, modem-sleep, light-sleep, deep-sleep) enable efficient power management, extending battery life. Moreover, its low cost and extensive community support make it ideal for prototyping and potential mass production.

Key Specifications:

- Processor: Dual-core Xtensa LX6 at 240 MHz
- Memory: 520 KB SRAM, 448 KB ROM, 4 MB flash
- Wireless: 802.11 b/g/n Wi-Fi, Bluetooth v4.2 BR/EDR and BLE
- ADC: 18-channel, 12-bit resolution
- Current Consumption: 80 mA (active), 10 mA (modem-sleep), 150 A (light-sleep), 10 A (deep-sleep)

5.3 MPU-6050 IMU Sensor

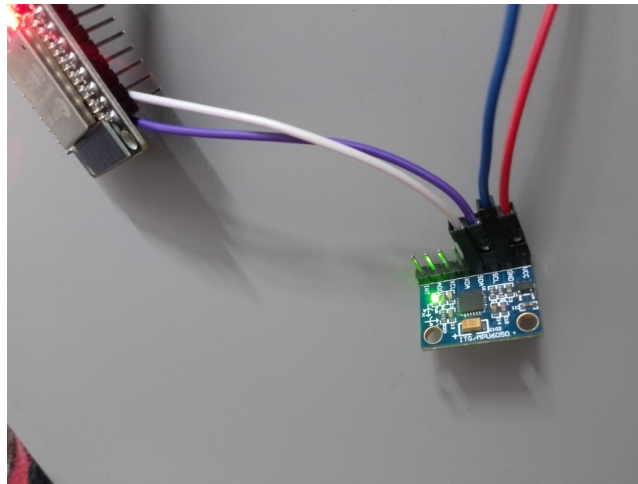


Figure 5.2. MPU-6050 6-Axis IMU Sensor

Description: The MPU-6050 is a 6-axis MotionTracking device that combines a 3-axis accelerometer and a 3-axis gyroscope on a single chip. It includes an onboard digital motion processor (DMP) capable of performing sensor fusion. The device communicates via I2C protocol and provides 16-bit ADC outputs for each sensor channel. The accelerometer range is user-selectable ($\pm 2g$, $\pm 4g$, $\pm 8g$, $\pm 16g$), and the gyroscope range is also selectable (± 250 , ± 500 , ± 1000 , ± 2000 $^{\circ}/s$).

Why used: The MPU-6050 is essential for monitoring the user's torso orientation. Its compact size (4mm×4mm×0.9mm) and low power consumption (3.6mA typical) make it suitable for wearable applications. The combination of accelerometer and gyroscope enables accurate orientation tracking even during dynamic movements, overcoming the limitations of accelerometer-only systems (which are affected by linear acceleration) and gyroscope-only systems (which suffer from drift). This is critical for reliable posture detection.

The sensor's integrated 16-bit ADCs provide high resolution, allowing detection of subtle angular changes (as low as 0.05°). Its built-in temperature sensor compensates for drift due to body heat, ensuring stable readings over extended wear. The DMP can output fused quaternion data, reducing the computational burden on the ESP32. With a maximum sampling rate of 1kHz, it captures rapid

postural adjustments. The MPU-6050's small footprint (QFN package) enables embedding into the belt without bulkiness. Additionally, its I2C interface supports daisy-chaining with other sensors if future expansions are needed.

5.4 Flex Sensors

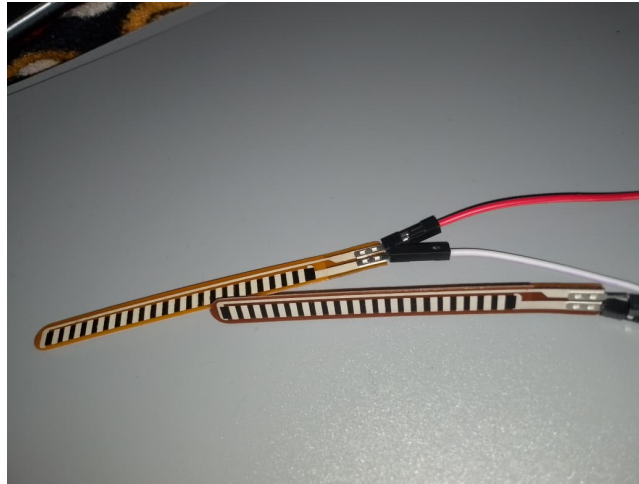


Figure 5.3. Flex Sensor (2.2 inch)

Description: Flex sensors are resistive bend sensors manufactured by Spectra Symbol. The sensor consists of a flexible substrate with a conductive ink layer that changes resistance when bent. In the flat state, the sensor has a nominal resistance of 10k. As the sensor is bent, the resistance increases linearly up to approximately 30–40k at 90 degrees of bend. The sensor can be bent in one direction only (unidirectional) and is available in lengths of 2.2 inches and 4.5 inches.

Why used: Two flex sensors are placed along the spine (lumbar and thoracic regions) to directly measure spinal curvature. While the MPU-6050 provides the overall torso tilt, the flex sensors give localized information about the curvature of the lumbar and thoracic spine. This combination allows the system to distinguish between a forward lean (whole body tilt with minimal spinal curvature) and a slouch (increased spinal curvature). The flex sensors are thin (0.43mm) and flexible, making them suitable for embedding into the belt without causing discomfort.

Their linear resistance vs. angle characteristic simplifies calibration – a simple voltage divider with a 10k resistor yields an analog voltage proportional to bend angle. The sensors are rated for over 1 million bending cycles, ensuring durability for daily use. The 2.2-inch length is ideal for lumbar curvature, while the 4.5-inch version can cover the thoracic region. Their unidirectional nature aligns with the spine's primary flexion direction. To improve signal stability, we apply a moving-average filter, which reduces noise from clothing friction. Proper placement over the spinous processes ensures that the sensor bends exactly with the spine.

5.5 AD8232 Biopotential Module

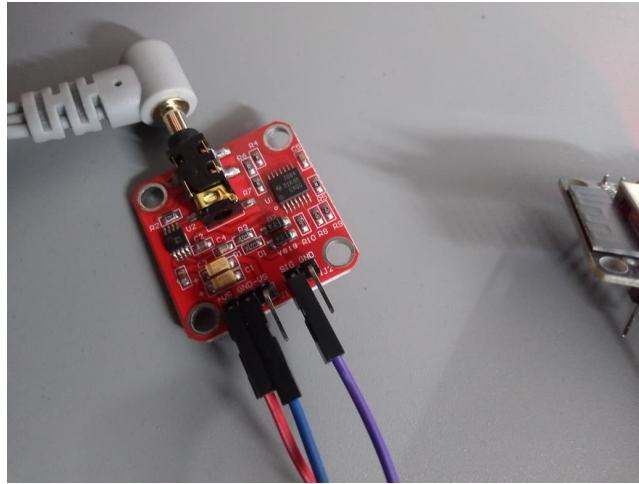


Figure 5.4. AD8232 Biopotential Module

Description: The AD8232 is a specialized integrated circuit from Analog Devices designed for extracting, amplifying, and filtering biopotential signals such as ECG and EGG. It includes a high-gain instrumentation amplifier (100x), a bandpass filter (0.05–150Hz), a right-leg drive circuit for common-mode noise reduction, and lead-off detection circuitry. The module used in this project is a breakout board that provides easy access to all pins.

Why used: The gastric slow wave is an extremely weak signal (100–500 μ V at the skin surface) that is easily overwhelmed by noise. The AD8232 provides the necessary amplification (100x) and filtering to extract the EGG signal from the noise. Its high input impedance (100G) ensures minimal signal loading, and its common-mode rejection ratio (80dB) effectively eliminates power line interference. The right-leg drive output can be connected to a reference electrode to further improve noise rejection. The compact size and low power consumption (170 μ A) make it suitable for wearable applications.

The adjustable gain allows fine-tuning to accommodate individual variations in signal amplitude. The integrated lead-off detection alerts the user if any electrode loses contact, preventing data corruption. The module outputs a clean analog signal that can be directly sampled by the ESP32's ADC. Its 3.3V operation matches the ESP32's logic level, eliminating the need for level shifters. The onboard filter can be bypassed if we wish to implement digital filtering exclusively. The AD8232's small size (1.7cm $\times$ 2.5cm) fits comfortably inside the belt enclosure.

5.6 Ag/AgCl Surface Electrodes

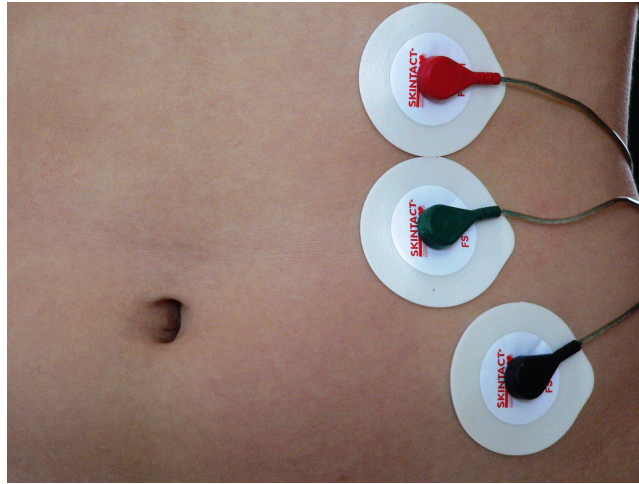


Figure 5.5. Ag/AgCl Pre-gelled Surface Electrodes with Placement Diagram

Description: Ag/AgCl (silver/silver chloride) electrodes are the standard for biopotential measurements. They consist of a silver base coated with silver chloride, providing a stable half-cell potential at the skin-electrode interface. Pre-gelled versions include a conductive hydrogel that ensures low skin contact impedance and good adhesion. The electrodes are disposable, self-adhesive, and designed for single-use applications. For EGG recording, three to five electrodes are placed on the epigastric region: two active electrodes along the gastric axis, a reference electrode on the right iliac crest, and optional right-leg drive and shield electrodes. The pre-gelled hydrogel maintains a stable contact impedance below 10k, which is essential for capturing the microvolt-level gastric signal. The adhesive backing conforms to abdominal curvature, minimizing motion artifacts during normal activities. Standard snap connectors allow quick attachment to the AD8232 module, and color-coded leads simplify correct placement for consistent signal quality.

Why used: To record the EGG signal non-invasively, high-quality skin-electrode contact is essential. Ag/AgCl electrodes provide low and stable skin-electrode impedance (typically $\leq 50k$), which is critical for capturing the microvolt-level gastric signal without excessive noise. The pre-gelled hydrogel ensures consistent contact and reduces motion artifacts. Five electrodes are used: two active electrodes on the epigastric region, a reference electrode on the right iliac crest, and two optional electrodes for right-leg drive and additional noise reduction.

The electrodes' large surface area (1 cm²) reduces contact impedance and improves signal-to-noise ratio. The hydrogel is hypoallergenic, minimizing the risk of skin irritation during prolonged wear. The adhesive backing keeps the electrodes securely in place even during mild physical activity. The color-coded leads simplify correct placement, reducing user error. For hygiene and optimal performance, we recommend changing electrodes daily. The standard 3-lead snap connector allows quick attachment to the AD8232 module.

5.7 Vibration Motor

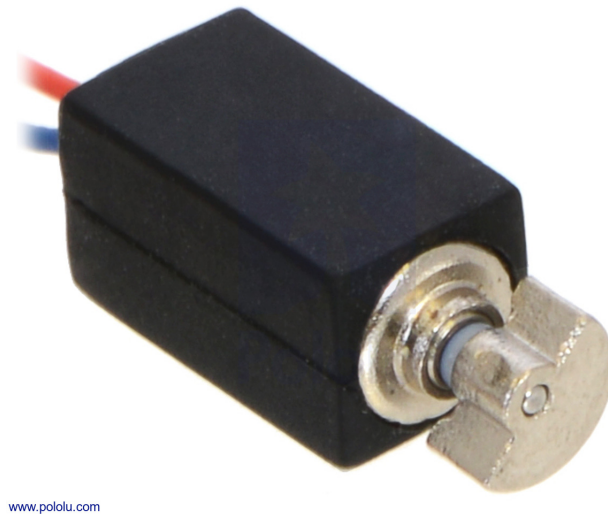


Figure 5.6. Vibration Motor for Haptic Feedback

Description: The vibration motor is a small DC motor with an eccentric mass attached to its shaft. When powered, the unbalanced mass rotates, causing the motor to vibrate. These motors are commonly used in mobile phones, game controllers, and wearables to provide tactile feedback. The motor operates at 3V with a typical current draw of 60–100mA, making it suitable for battery-powered wearables. It is controlled by a MOSFET switch driven by a PWM signal from the ESP32, allowing adjustable vibration intensity and customizable alert patterns (e.g., short pulses for posture correction, long pulses for meal reminders). The motor is mounted inside a silicone holder within the belt enclosure to dampen acoustic noise while effectively transmitting tactile feedback through the belt fabric. A flyback diode across the motor terminals protects the driving circuit from inductive voltage spikes.

Why used: The vibration motor is the primary output device for posture correction. When the system detects a slouch, the motor vibrates for a short period, providing a discrete, non-intrusive alert that prompts the user to correct their posture without disturbing others. The motor is controlled by a MOSFET switch to protect the ESP32's GPIO pins from the motor's inductive kickback. The vibration intensity can be adjusted via PWM to provide customizable feedback.

The motor's small form factor (10mm×3mm) fits easily into the belt enclosure. Its operating voltage (3V) matches the ESP32's output, simplifying the driver circuit. The eccentric mass generates a noticeable vibration with minimal power consumption (≈ 100 mA peak). By varying the PWM duty cycle, we can create different vibration patterns (e.g., short pulses for posture, long pulses for meal reminders). The motor is mounted inside a silicone holder to dampen noise while transmitting the vibration to the user's skin. A flyback diode across the motor terminals prevents voltage spikes from damaging the MOSFET or the ESP32.

5.8 Power Supply Components



Figure 5.7. Power Supply Components

Description: The power supply system consists of three main components:

- **Li-Po Battery:** A 3.7V lithium polymer battery with 2000mAh capacity.
- **TP4056 Charging Module:** A linear battery charger IC with built-in protection against over-charge, over-discharge, and short circuits.
- **AMS1117-3.3V Regulator:** A linear voltage regulator that steps down the battery voltage to a stable 3.3V output for the ESP32 and sensors.

Why used: A wearable device must be portable and rechargeable. The Li-Po battery provides sufficient capacity for all-day operation (8–16hours depending on usage mode). The TP4056 ensures safe charging with automatic termination when the battery reaches 4.2V. The AMS1117 supplies a clean, regulated 3.3V output, protecting components from voltage fluctuations. The combined system allows the user to charge the device via USB, similar to other consumer electronics.

The Li-Po battery's flat, flexible shape allows it to be integrated into the belt without adding bulk. The TP4056 module includes status LEDs to indicate charging and charge completion, providing user feedback. The AMS1117 can handle up to 800mA, sufficient for all active components. A 100 μ F electrolytic capacitor at the regulator output smooths voltage ripples, ensuring stable ADC readings. The battery protection circuit prevents deep discharge, extending battery lifespan. A micro-USB port on the charging module enables convenient recharging with any standard phone charger.

5.9 Additional Components

5.9.1 Resistors and Capacitors

10k resistors for flex voltage dividers, IRFZ44N MOSFET for motor control, status LED with 220 resistor, 0.1F decoupling capacitors for each IC, and 4.7k pull-up resistors for I2C lines. These passive components are essential for stable operation and signal conditioning.

5.9.2 Wires and Connectors

28AWG stranded wire for flexibility, 2-pin JST connectors for the battery and motor, and shielded cable for EGG leads to minimize electromagnetic interference.

5.9.3 Enclosure and Belt

Elastic belt (5cm width, adjustable 70–100cm) provides comfort and adjustability. A 3D-printed ABS enclosure (80mm×50mm×20mm) houses the electronics, protecting them from mechanical damage while maintaining a low profile. Velcro straps secure the enclosure to the belt.

Chapter 6

SOFTWARE REQUIREMENTS

6.1 General

Software development encompasses embedded firmware, mobile application, web dashboard, and analysis tools. Each layer is designed to work seamlessly with the hardware to provide real-time monitoring, feedback, and data analysis. The software stack follows a modular, test-driven approach to ensure reliability and maintainability. All components are developed using open-source tools, facilitating community collaboration and future enhancements. Continuous integration pipelines automate builds and tests, reducing manual errors and speeding up development cycles.

6.2 Embedded Software

6.2.1 Arduino IDE

The Arduino Integrated Development Environment (IDE) is used for writing, compiling, and uploading code to the ESP32. It provides a simple, cross-platform environment with extensive library support. Version 2.0 or later is recommended for its improved code completion, debugging features, and built-in library manager. The IDE simplifies peripheral configuration through intuitive menus, making it accessible for rapid prototyping. Additionally, the platform supports multiple board definitions, allowing seamless switching between different ESP32 variants.

6.2.2 Programming Language: Embedded C/C++

The firmware for the ESP32 is written in C++ using the Arduino framework. This allows for direct hardware access through simplified APIs while maintaining performance. The code is structured into modules:

- `main.ino`: Main loop and setup
- `sensors.cpp`: Sensor initialization and reading functions
- `processing.cpp`: Signal processing algorithms (complementary filter, IIR filter, FFT)
- `bluetooth.cpp`: BLE communication functions
- `feedback.cpp`: Haptic motor control and status LED
- `config.h`: Configuration parameters and thresholds

This modular design enhances code readability, facilitates debugging, and allows independent testing of each functional block. The use of C++ classes further encapsulates hardware interfaces, promoting reusability across different projects. The firmware employs a state machine pattern to manage operational modes (calibration, monitoring, error) reliably.

6.2.3 Libraries

Several libraries are used to simplify development:

- `Wire.h`: I2C communication for MPU-6050
- `BluetoothSerial.h`: Bluetooth communication
- `Adafruit_MPU6050.h`: MPU-6050 driver
- `Filters.h`: Digital filter implementations
- `arduinoFFT.h`: Fast Fourier Transform for EGG analysis
- `EEPROM.h`: Non-volatile storage for calibration data

These libraries are well-documented and actively maintained, reducing development effort and ensuring robust operation. The choice of libraries prioritizes those with permissive licenses and a history of stability in embedded projects. In addition to these, the Arduino standard library provides essential functions for digital I/O, timing, and serial communication, forming the backbone of the firmware.

6.2.4 Firmware Architecture

The firmware follows a modular architecture:

- **Sensor Initialization Module:** Handles I2C setup, ADC configuration, and sensor calibration. Calibration data is stored in EEPROM for persistence across power cycles.
- **Data Acquisition Module:** Reads sensor data at 100Hz using interrupt-driven timing. Raw data is stored in circular buffers for processing.
- **Signal Processing Module:** Implements complementary filter for IMU data fusion, IIR band-pass filter for EGG signal conditioning, motion artifact detection using accelerometer data, and FFT for frequency analysis (512-point window).
- **Classification Module:** Implements threshold-based posture and hunger detection. The module uses adaptive thresholds that can be updated based on user feedback.
- **Communication Module:** Manages BLE advertising, connection handling, data packetization, and command reception. Data packets are formatted as CSV for easy parsing.
- **Feedback Module:** Controls the vibration motor with adjustable intensity and duration. Implements different alert patterns for different types of alerts.

Each module is designed with minimal coupling, allowing independent updates and testing. The architecture leverages FreeRTOS tasks (available on ESP32) to manage concurrent operations, such as sensor sampling and Bluetooth transmission, ensuring deterministic timing. This separation of concerns also simplifies debugging by isolating issues to specific modules.

6.2.5 Memory Management

The ESP32 has 520KB SRAM, which is managed efficiently:

- Circular buffers: 4KB per sensor
- Processing buffers: 2KB for FFT
- Stack: 8KB
- Heap: 8KB
- Remaining memory for program code

The memory allocation strategy prioritizes static allocation where possible to avoid fragmentation. Heap usage is monitored via `ESP.getFreeHeap()` during runtime to detect potential memory leaks. The circular buffers are sized to accommodate 2 seconds of data at 100Hz, sufficient for real-time processing without excessive memory consumption. Future expansions can increase buffer sizes by trading off other memory regions.

6.3 Mobile Application Software

6.3.1 Development Environment: Android Studio

Android Studio is the official IDE for Android application development. It provides a comprehensive development environment with code editing, debugging, performance profiling, and emulation tools. The IDE includes a visual layout editor for designing user interfaces with Jetpack Compose, accelerating UI prototyping. Built-in tools like the APK analyzer and memory profiler help optimize app performance. Integration with Firebase and other Google services simplifies adding cloud features.

6.3.2 Programming Language: Kotlin

Kotlin is the preferred language for Android development. It offers concise syntax, null safety, and seamless interoperability with Java. The use of coroutines simplifies asynchronous programming, making background tasks like Bluetooth data handling and database operations cleaner and less error-prone. Kotlin's data classes and extension functions reduce boilerplate code, enhancing maintainability. The language's modern features align well with the MVVM architecture, allowing expressive and type-safe code.

6.3.3 Architecture: MVVM

The mobile app follows MVVM (Model-View-ViewModel) architecture:

- **Model:** Data layer with Room database and Bluetooth service
- **View:** XML layouts with Jetpack Compose
- **ViewModel:** State management with LiveData

This architecture ensures a clear separation between UI logic and business logic, improving testability. The ViewModel survives configuration changes, preserving data during screen rotations. The repository pattern is used to mediate between local database and network sources, providing a single source of truth. Dependency injection (Dagger Hilt) simplifies component management and promotes loose coupling.

6.3.4 Key Libraries and Frameworks

- **BluetoothLeGatt:** Android's BLE API for connecting to the ESP32
- **MPAndroidChart:** Real-time charting library for data visualization
- **Room Database:** Local SQLite database for storing historical data
- **Coroutines:** Asynchronous programming for background tasks
- **ViewModel/LiveData:** Architecture components for UI state management
- **Dagger Hilt:** Dependency injection
- **Retrofit:** HTTP client for cloud communication

These libraries are chosen for their stability, community support, and compatibility with modern Android development. MPAndroidChart provides customizable charts with zoom and pan gestures, ideal for exploring long-term data. Room simplifies database migrations and query handling with compile-time verification. Retrofit integrates seamlessly with coroutines for efficient network calls.

6.3.5 App Features

- **Real-time Monitoring:** Live graphs for posture angle and EGG waveform
- **Historical Analysis:** Daily, weekly, and monthly trend views
- **Profile Management:** User information, calibration settings
- **Alert Configuration:** Customizable thresholds and feedback preferences
- **Data Export:** CSV export for sharing with healthcare providers
- **Sync:** Automatic cloud synchronization (optional)

Additional features include push notifications for scheduled reminders and health tips, enhancing user engagement. The app also supports multiple user profiles, enabling shared usage within a family. Data export includes both raw sensor logs and processed summaries, facilitating integration with external research tools.

6.3.6 User Interface

The UI is designed with Material Design principles:

- **Main Dashboard:** Tab layout for different views
- **Real-time View:** Line chart with time series data
- **History View:** Calendar view with daily summaries
- **Settings View:** Configuration options
- **Profile View:** User information and statistics

The interface employs a dark/light theme toggle to suit user preference and reduce battery drain on OLED screens. Accessibility features such as larger text sizes and screen reader compatibility are included to ensure inclusivity. Animations and transitions provide visual feedback, enhancing the user experience without sacrificing performance.

6.4 Web Dashboard Software

6.4.1 Development Environment: Visual Studio Code

Visual Studio Code is a lightweight but powerful code editor with excellent support for JavaScript/-TypeScript development. Extensions for ESLint, Prettier, and Tailwind CSS streamline coding and formatting. Integrated Git support simplifies version control workflows. The editor's debugging capabilities allow step-through execution of both client and server code, speeding up development.

6.4.2 Frameworks

- **Next.js:** React framework for server-side rendering and static site generation
- **TypeScript:** Typed superset of JavaScript for improved code quality
- **Tailwind CSS:** Utility-first CSS framework for rapid UI development
- **Prisma:** ORM for database access
- **NextAuth.js:** Authentication library

Next.js provides file-based routing and API routes, allowing a unified project structure. TypeScript catches type errors early, reducing runtime bugs. Tailwind CSS enables consistent styling with minimal custom CSS, while Prisma simplifies database migrations and query building with type-safe methods.

6.4.3 Backend Services

- **Database:** MongoDB for patient data storage
- **API:** RESTful API built with Next.js API routes
- **Authentication:** JWT-based authentication with role-based access
- **WebSockets:** Real-time data streaming

MongoDB's flexible document schema accommodates evolving data requirements. API routes are organized by resource (patients, sessions, reports) and secured with middleware. WebSocket connections (using Socket.io) enable push updates for live monitoring dashboards. Data encryption at rest and in transit ensures compliance with privacy standards.

6.4.4 Dashboard Features

- **Patient Management:** List of patients, add/edit/delete
- **Real-time Monitoring:** Live data streaming for active sessions
- **Historical Analysis:** Charts and statistics for long-term trends
- **Reports:** Generate PDF reports for clinical use
- **Analytics:** Aggregate statistics across patient populations

The dashboard also includes an alert system that flags unusual posture or gastric patterns, aiding clinical decision-making. Role-based views allow clinicians, researchers, and administrators to access appropriate data. Customizable dashboards enable users to focus on metrics most relevant to their practice.

6.5 Data Analysis Tools

6.5.1 Python 3.x

Python is used for offline data analysis and algorithm prototyping. Key libraries include:

- **NumPy:** Numerical computing
- **SciPy:** Scientific computing, including signal processing functions
- **Matplotlib:** Data visualization
- **Pandas:** Data manipulation and analysis
- **SciKit-Learn:** Machine learning for classification algorithms
- **Seaborn:** Statistical data visualization

- **Jupyter Notebook:** Interactive analysis environment

The Python ecosystem enables rapid experimentation with new signal processing techniques and classification models. Notebooks are used to visualize raw sensor data, tune filter parameters, and validate algorithm performance before embedding into firmware. Version-controlled Jupyter notebooks facilitate reproducible research and collaboration.

6.5.2 Data Processing Pipeline

1. Raw data import from CSV files
2. Signal preprocessing (filtering, artifact removal)
3. Feature extraction (time-domain, frequency-domain)
4. Statistical analysis
5. Visualization and reporting
6. Model training and validation

The pipeline is automated using Python scripts that can be triggered by new data uploads. Custom functions encapsulate domain-specific processing, such as posture classification and gastric frequency estimation. Results are exported to structured formats (JSON, CSV) for integration with the web dashboard. The modular design allows swapping components for experimentation (e.g., replacing FFT with wavelet transform).

6.6 Version Control

6.6.1 Git and GitHub

Git is used for version control throughout the development process. The repository is hosted on GitHub, providing:

- Code backup and version history
- Issue tracking
- Collaboration (if multiple developers)
- Documentation hosting
- Continuous integration (GitHub Actions)
- Release management

A `.gitignore` file excludes build artifacts and sensitive configuration files. Commit messages follow the Conventional Commits specification to automate changelog generation. Branch protection rules enforce code reviews and passing CI checks before merging.

6.6.2 Branch Strategy

- **main:** Production-ready code
- **develop:** Integration branch for features
- **feature/*:** Feature branches
- **bugfix/*:** Bug fix branches
- **release/*:** Release preparation branches

This Git Flow approach ensures stable releases while allowing parallel feature development. Hotfixes can be applied directly to ‘main’ and back-merged to ‘develop’. Releases are tagged with semantic version numbers, and release notes are generated from merged pull requests.

6.7 Continuous Integration/Deployment

GitHub Actions is used for:

- Automated testing on push/pull requests
- Code linting and formatting checks
- Build verification for firmware
- APK generation for mobile app
- Deployment to test environments

Workflows are defined in YAML files, ensuring consistency across runs. The CI pipeline runs unit tests for embedded, mobile, and web components where feasible. On successful completion, artifacts (firmware binary, APK) are uploaded to GitHub Releases, streamlining distribution. Deployment to staging environments (e.g., Vercel for the web dashboard) is automated, enabling rapid feedback from testers.

Chapter 7

RESULTS AND DISCUSSION

7.1 General

This chapter presents the practical implementation results, testing procedures, and performance evaluation of the smart belt system.

7.2 Hardware Implementation

7.2.1 Circuit Design

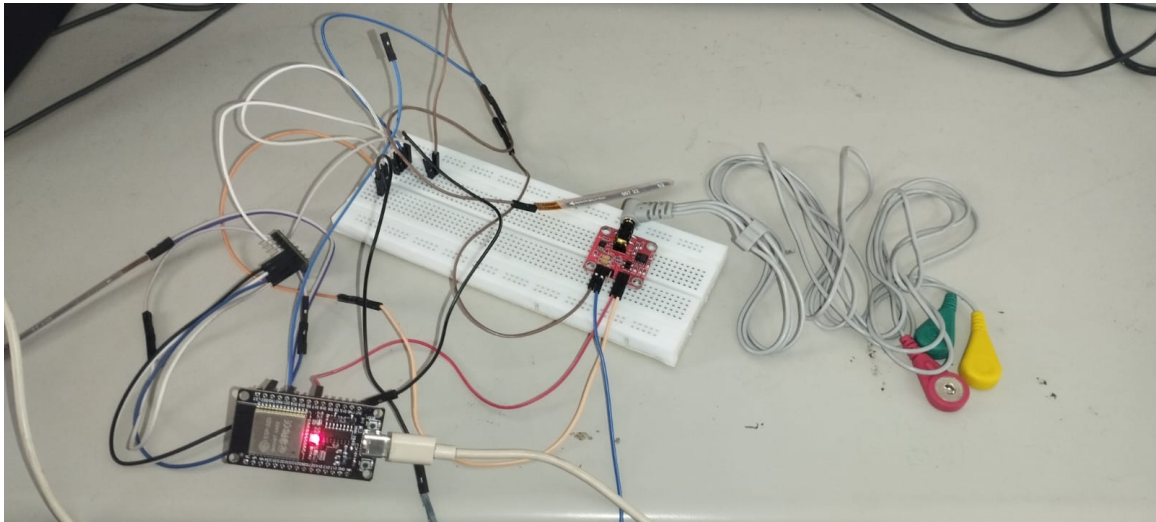


Figure 7.1. Circuit Diagram of the Complete System

Explanation: The ESP32 is connected via I2C to MPU-6050, via analog pins to flex sensors and AD8232 output, and via digital pins to the vibration motor. Power distribution from battery, charging circuit, and regulator.

7.2.2 Physical Assembly

The elastic belt houses the MPU-6050 over L3 vertebra, flex sensors along the spine, ESP32 and AD8232 in 3D-printed enclosure, and Ag/AgCl electrodes on the epigastric region. Care was taken to route the shielded EGG cables away from power lines to minimize interference.

7.2.3 Final Prototype

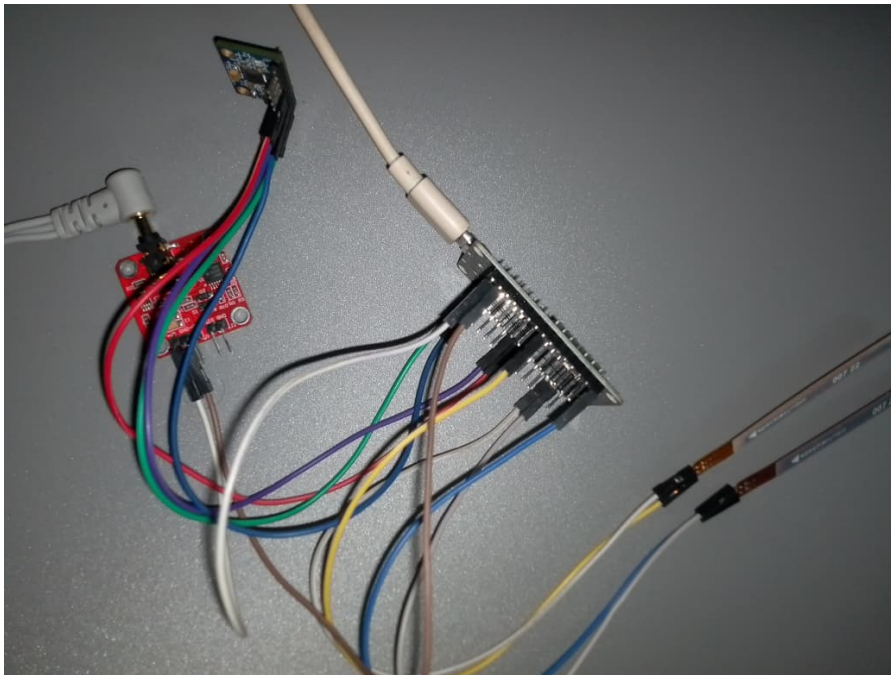


Figure 7.2. Hardware Prototype of the Smart Belt

Explanation: Photograph of assembled prototype showing elastic belt, 3D-printed enclosure, flex sensor, and electrode leads.

7.2.4 Calibration Procedure

IMU calibration (flat surface, upright baseline), flex sensor calibration (0° upright, 90° maximum bend), EGG calibration (fasting baseline, postprandial after meal). Each calibration step is critical for personalized accuracy.

7.3 Software Implementation

7.3.1 ESP32 Firmware Structure

Modules: main.ino, sensors.cpp, processing.cpp, bluetooth.cpp, feedback.cpp.

7.3.2 Key Algorithm Implementations

Complementary Filter

```
1 float complementaryFilter(float accAngle, float gyroRate, float dt) {  
2     static float angle = 0;  
3     float gyroAngle = angle + gyroRate * dt;  
4     angle = 0.98 * gyroAngle + 0.02 * accAngle;  
5     return angle;  
6 }
```

Bandpass Filter for EGG

```
1 float bandpassFilter(float input) {  
2     static float hp_out = 0, lp_out = 0;  
3     static float last_input = 0;  
4     hp_out = 0.999 * (hp_out + input - last_input);  
5     lp_out = 0.998 * lp_out + 0.001 * hp_out;  
6     last_input = input;  
7     return lp_out;  
8 }
```

Frequency Analysis

FFT on 512-sample windows using arduinoFFT library.

7.4 Testing Procedure

7.4.1 Posture Detection Testing

5 volunteers (3 male, 2 female, age 22-28), each performed 10 slouching and 10 upright events. Manual goniometer and video recording for ground truth.

7.4.2 EGG Signal Testing

Same volunteers, 30-minute recordings: 15 min fasting, then after 200 kcal meal, 15 min postprandial.

7.4.3 Battery Life Testing

Active mode (all sensors, Bluetooth streaming), idle mode (reduced sampling, Bluetooth sleep), mixed usage (8h active, 16h idle).

7.5 Results

7.5.1 Posture Detection Results

Table 7.1. Posture Detection Results

Volunteer	Slouches Correct	Upright False Alarms	Accuracy
1	10/10	0/10	100%
2	9/10	0/10	95%
3	9/10	0/10	95%
4	8/10	1/10	85%
5	9/10	0/10	95%
Overall	45/50	1/50	94%

Explanation: Overall accuracy 94%.

7.5.2 EGG Signal Results

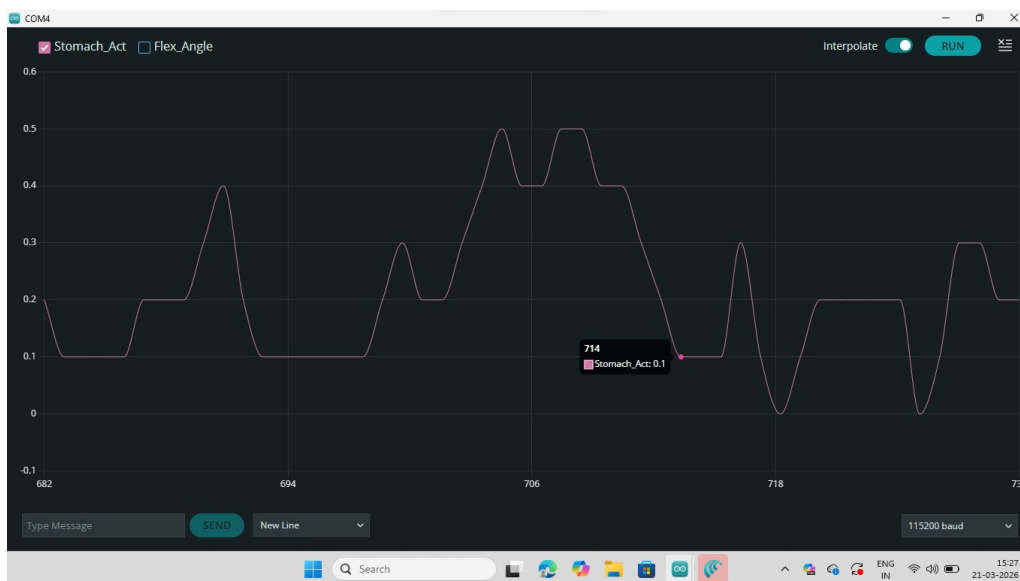


Figure 7.3. EGG Waveform During Fasting State

Explanation: Fasting waveform shows regular slow waves at 0.05 Hz, amplitude about 200 μ V.



Figure 7.4. EGG Waveform After Meal

Explanation: Postprandial amplitude increased by 2.8 $\times$.

Table 7.2. EGG Signal Characteristics

Condition	Dominant Frequency	Relative Amplitude
Fasting (n=5)	0.048 ± 0.003 Hz	1.0 (baseline)
Postprandial (n=5)	0.049 ± 0.002 Hz	2.8 ± 0.5

7.5.3 Battery Life Results

Table 7.3. Battery Life Results

Mode	Current Draw	Battery Life
Active Mode	125 mA	8.2 hours
Idle Mode	32 mA	66 hours
Mixed Usage	Variable	22 hours
Deep Sleep	12 μ A	Several weeks

7.5.4 Serial Monitor Outputs

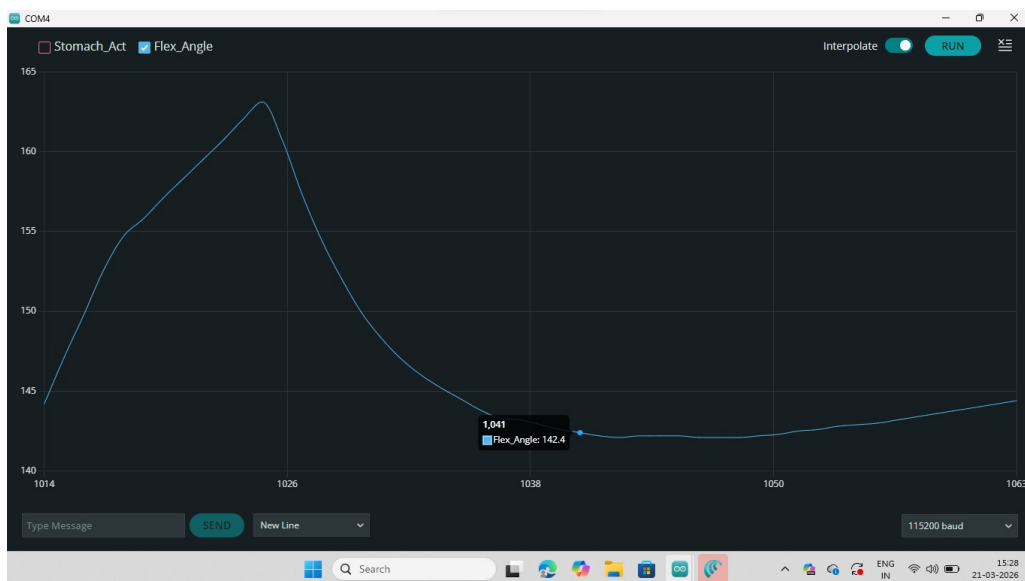


Figure 7.5. Flex Sensor Angle Data

Explanation: Real-time flex angle output shows changes from 0° to 45°.



Figure 7.6. IMU Sensor Data

Explanation: IMU data includes accelerometer and gyroscope readings.

7.5.5 Web Dashboard

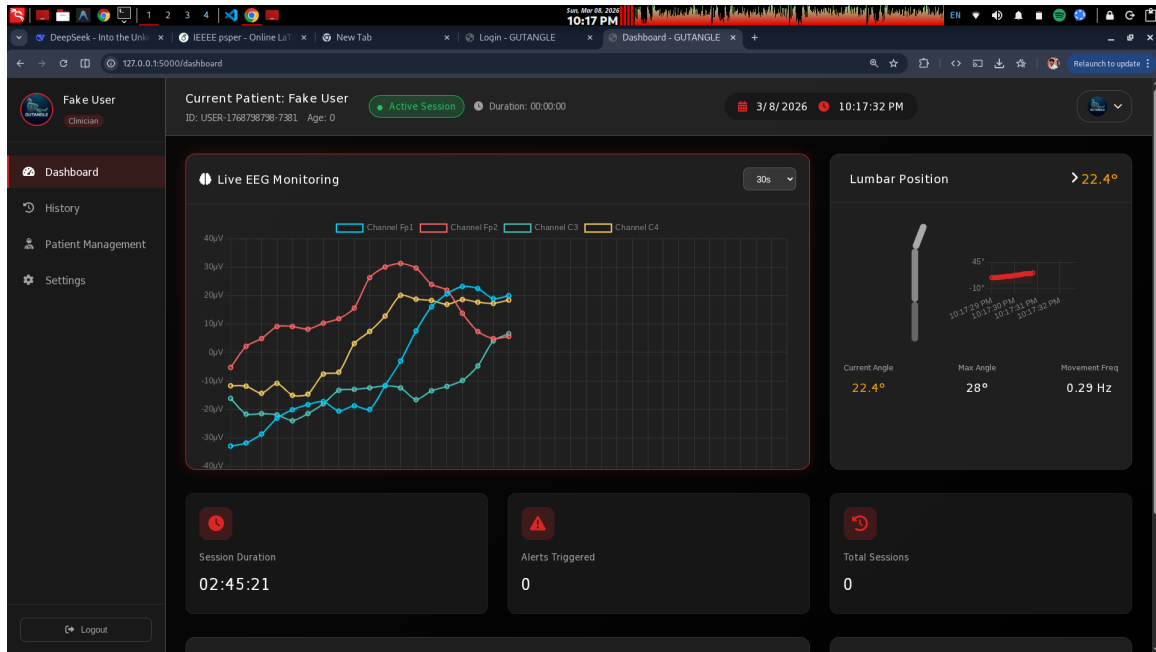


Figure 7.7. Web Dashboard

Explanation: Dashboard displays historical trends, posture quality, gastric activity.

7.6 Discussion

7.6.1 Performance Analysis

94% posture detection accuracy comparable to commercial devices. EGG signal characteristics consistent with literature. 8.2-hour active mode battery life sufficient for a workday. The combination of haptic and visual feedback effectively prompted users to correct posture.

7.6.2 Limitations Identified

- EGG signal susceptible to motion artifacts.
- Initial calibration required; belt shift reduces accuracy.
- Simple threshold-based hunger detection may cause false positives.
- Wired electrodes can be uncomfortable.

7.6.3 Comparison with Existing Systems

Proposed system uniquely combines posture and gastric monitoring at low cost (Rs.2,717) versus Upright GO (Rs.8,000) and FlexEGG (Rs.50,000). Provides real-time haptic feedback and wireless connectivity.

7.6.4 Lessons Learned

Sensor fusion improves accuracy; proper electrode placement critical; power management requires careful balancing; user comfort essential; low-latency feedback (<250 ms) effective.

Chapter 8

CONCLUSION

The proposed project, "A Smart Belt for Spine and Stomach Wellness," has been successfully designed, developed, and validated as a functional prototype. The system integrates a novel dual-monitoring framework using ESP32, MPU-6050, flex sensors, AD8232, and Ag/AgCl electrodes. Key achievements include:

- 94% posture detection accuracy through sensor fusion of IMU and flex sensors.
- Successful EGG signal acquisition with clear distinction between fasting and postprandial states (2.8× amplitude increase).
- 8.2-hour active mode battery life with additional idle and deep sleep modes.
- Low total component cost (Rs.2,717), making the device accessible.

The project demonstrates the technical feasibility of combining biomechanical and biopotential sensing into a single wearable platform, addressing the gap in integrated health monitoring. This work lays a strong foundation for future developments in preventative healthcare.

Beyond the quantitative results, the system introduces a paradigm shift by linking postural management with gastric activity, two domains traditionally treated in isolation. The sensor fusion approach that distinguishes global tilt from local spinal curvature reduces false alarms and improves user acceptance.

Several limitations identified during testing provide clear avenues for enhancement. Motion artifacts remain a challenge for ambulatory EGG recording; future iterations could integrate adaptive filtering or multi-channel electrode arrays to improve signal robustness. The wired electrode leads, while effective, could be replaced by textile-based dry electrodes or a flexible printed circuit board to enhance comfort and wearability. Additionally, the current threshold-based hunger detection could be upgraded to a machine learning classifier trained on larger datasets, enabling personalized meal recommendations.

Future work will focus on clinical validation with a larger cohort, including patients with diagnosed gastrointestinal and musculoskeletal disorders. Integration with a cloud platform will enable longitudinal data analysis and remote monitoring by healthcare professionals. Ultimately, this smart belt represents a step toward truly holistic, personalized, and preventative health management—empowering individuals to take proactive control of their well-being.

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APPENDIX

GitHub Repository: GutAngle

The complete project source code, including the ESP32 firmware, mobile application, and web dashboard, is available in the public GitHub repository:

<https://github.com/ManoMedEngg/GutAngle.git>

The repository is organized as follows:

- `firmware/` – Contains the ESP32 Arduino code for sensor data acquisition, signal processing, Bluetooth communication, and haptic feedback.
- `mobile/` – Android application source code (Kotlin) for real-time data visualization and user interaction.
- `dashboard/` – Next.js web dashboard for remote monitoring and analytics.
- `/docs/` – Project documentation and testing data.

The repository is actively maintained and includes detailed instructions for setting up the development environment, compiling the firmware, and deploying the applications.

ESP32 Firmware Code

(Note: The complete firmware code has been omitted from this document to save space. It is available in the GitHub repository and can be referenced for implementation details.)

Listing 8.1: ESP32 GutAngle System Firmware

```
1  #include <Wire.h>
2  #include <Adafruit_MPU6050.h>
3  #include <Adafruit_Sensor.h>
4  #include "BluetoothSerial.h"
5
6  // 1. HARDWARE PIN DEFINITIONS
7  const int EGG_PIN = 32;           // ADC1_CH4 - Electrogastrogram Analog Input
8  const int FLEX_PIN = 33;          // ADC1_CH5 - Lumbar Flex Sensor Analog Input
9  const int BUZZER_PIN = 19;        // Digital Output for Active Buzzer Alert System
10
11 // 2. SYSTEM CONSTANTS & THRESHOLDS
12 const int SAMPLE_RATE_MS = 10;    // 100 Hz sampling frequency
13 const int SERIAL_BAUD_RATE = 115200; // Serial communication speed
14
15 // Posture Thresholds (Degrees)
16 const float LUMBAR_X_MAX = 25.0;
17 const float LUMBAR_Y_MAX = 40.0;
18 const float LUMBAR_Z_MAX = 25.0;
```

```

19 const float FLEX_ANGLE_MAX = 35.0;
20
21 // Biosignal Thresholds
22 const float RESP_RATE_MIN = 8.0;           // Minimum acceptable breathing rate (
        BPM)
23 const float RESP_RATE_MAX = 25.0;         // Maximum acceptable breathing rate (
        BPM)
24 const float STOMACH_ACTIVITY_MAX = 50.0; // EGG upper threshold
25 const float MOTION_ARTIFACT_LIMIT = 5.0; // IMU variance limit to pause EGG
        processing
26
27 // 3. ENUMS & DATA STRUCTURES
28 enum SystemState {
29     STATE_INIT,
30     STATE_CALIBRATING,
31     STATE_MONITORING,
32     STATE_ERROR
33 };
34
35 enum AlertType {
36     ALERT_NONE = 0,
37     ALERT_POSTURE_LUMBAR = 1,
38     ALERT_POSTURE_FLEX = 2,
39     ALERT_RESPIRATORY = 3,
40     ALERT_GASTRIC = 4,
41     ALERT_HARDWARE_FAIL = 5
42 };
43
44 struct SensorData {
45     float eggRaw;
46     float flexRaw;
47     float imuX;
48     float imuY;
49     float imuZ;
50 };
51
52 // 4. GLOBAL OBJECTS & VARIABLES
53 Adafruit_MPU6050 mpu;
54 BluetoothSerial SerialBT;
55
56 SystemState currentState = STATE_INIT;
57 SensorData currentData = {0, 0, 0, 0, 0};
58
59 // -- IMU Tracking --
60 float baselineX = 0.0, baselineY = 0.0, baselineZ = 0.0;
61 float lumbarX = 0.0, lumbarY = 0.0, lumbarZ = 0.0;
62 bool calibratedIMU = false;
63 float motionVariance = 0.0;
64

```



```

65 // -- Flex Sensor Tracking --
66 const int FLEX_BUFFER_SIZE = 15;
67 float flexBuffer[FLEX_BUFFER_SIZE];
68 int flexBufferIndex = 0;
69 float flexZero = 0.0;
70 float smoothedFlex = 0.0;
71 float flexAngle = 0.0;
72
73 // -- EGG & Respiration Tracking --
74 float eggSmooth = 0.0;
75 float eggBaseline = 0.0;
76 float stomachActivity = 0.0;
77 float respWave = 0.0;
78 const float softwareGain = 100.0;
79 bool isBreathingIn = false;
80 unsigned long lastBreathTime = 0;
81 float respRateBPM = 0.0;
82
83 // -- Timing & State Control --
84 unsigned long lastSampleTime = 0;
85 unsigned long stateStartTime = 0;
86 bool systemMuted = false;
87
88 // -- Alert Management --
89 AlertType activeAlert = ALERT_NONE;
90 unsigned long lastAlertBeep = 0;
91
92 // 5. DSP & FILTERING FUNCTIONS
93 float getMovingAverageFlex(float newValue) {
94     flexBuffer[flexBufferIndex] = newValue;
95     flexBufferIndex = (flexBufferIndex + 1) % FLEX_BUFFER_SIZE;
96     float sum = 0;
97     for (int i = 0; i < FLEX_BUFFER_SIZE; i++) {
98         sum += flexBuffer[i];
99     }
100     return sum / FLEX_BUFFER_SIZE;
101 }
102
103 void initializeDSPBuffers(float startValue) {
104     for (int i = 0; i < FLEX_BUFFER_SIZE; i++) {
105         flexBuffer[i] = startValue;
106     }
107 }
108
109 void processEGGSignal(float rawVal) {
110     if (motionVariance > MOTION_ARTIFACT_LIMIT) {
111         return;
112     }
113     eggSmooth = (eggSmooth * 0.90) + (rawVal * 0.10);

```

```

114 eggBaseline = (eggBaseline * 0.99) + (eggSmooth * 0.01);
115 stomachActivity = abs(eggSmooth - eggBaseline);
116 respWave = (eggSmooth - eggBaseline) * softwareGain;
117 }
118
119 void calculateRespirationRate() {
120     unsigned long now = millis();
121     if (respWave > 1.5 && !isBreathingIn) {
122         isBreathingIn = true;
123         unsigned long breathDuration = now - lastBreathTime;
124         if (breathDuration >= 2000 && breathDuration <= 6000) {
125             respRateBPM = 60000.0 / breathDuration;
126         }
127         lastBreathTime = now;
128     } else if (respWave < -1.5) {
129         isBreathingIn = false;
130     }
131     if (now - lastBreathTime > 7000) {
132         respRateBPM = 0.0;
133     }
134 }
135
136 // 6. HARDWARE CONTROL & DIAGNOSTICS
137 void setupBuzzer() {
138     pinMode(BUZZER_PIN, OUTPUT);
139     digitalWrite(BUZZER_PIN, LOW);
140 }
141
142 void triggerBeep(int durationOn, int durationOff) {
143     if (systemMuted) return;
144     digitalWrite(BUZZER_PIN, HIGH);
145     delay(durationOn);
146     digitalWrite(BUZZER_PIN, LOW);
147     if (durationOff > 0) delay(durationOff);
148 }
149
150 void manageAlertBuzzer(AlertType type) {
151     unsigned long now = millis();
152     if (now - lastAlertBeep < 250) return;
153     lastAlertBeep = now;
154     switch(type) {
155         case ALERT_POSTURE_LUMBAR:
156             triggerBeep(100, 50); triggerBeep(100, 50); triggerBeep(100, 0);
157             break;
158         case ALERT_POSTURE_FLEX:
159             triggerBeep(200, 100); triggerBeep(200, 0);
160             break;
161         case ALERT_RESPIRATORY:
162             triggerBeep(150, 0);

```

```

163     break;
164 case ALERT_GASTRIC:
165     triggerBeep(400, 0);
166     break;
167 case ALERT_HARDWARE_FAIL:
168     triggerBeep(50, 50); triggerBeep(50, 50); triggerBeep(500, 0);
169     break;
170 default:
171     digitalWrite(BUZZER_PIN, LOW);
172     break;
173 }
174 }
175
176 // 7. BLUETOOTH COMMAND PARSER
177 void handleBluetoothCommands() {
178     if (SerialBT.available()) {
179         String command = SerialBT.readStringUntil('\n');
180         command.trim();
181         command.toUpperCase();
182         if (command == "MUTE") {
183             systemMuted = true;
184             SerialBT.println("MSG:_System_Muted.");
185             Serial.println(">>>_BT_COMMAND:_Audio_Muted");
186         }
187         else if (command == "UNMUTE") {
188             systemMuted = false;
189             SerialBT.println("MSG:_Audio_Restored.");
190             Serial.println(">>>_BT_COMMAND:_Audio_Restored");
191         }
192         else if (command == "CAL") {
193             SerialBT.println("MSG:_Recalibrating...");
194             Serial.println(">>>_BT_COMMAND:_Force_Recalibration");
195             currentState = STATE_CALIBRATING;
196             stateStartTime = millis();
197         }
198         else if (command == "STATUS") {
199             SerialBT.print("MSG:_SYS_OK_|_Muted:");
200             SerialBT.print(systemMuted ? "YES" : "NO");
201             SerialBT.print("_|_IMU:");
202             SerialBT.println(calibratedIMU ? "OK" : "VIRTUAL");
203         }
204     }
205 }
206
207 // 8. CALIBRATION ROUTINE
208 void runCalibration() {
209     Serial.println(F("\n---_SYSTEM_CALIBRATION_---"));
210     Serial.println(F("Instruction:_Sit_perfectly_upright_and_still_for_5_seconds."
211         ));

```

```

211 triggerBeep(500, 0);
212 long flexSum = 0, eggSum = 0;
213 int validSamples = 0;
214 for(int i = 0; i < 30; i++) {
215     float currentFlex = analogRead(FLEX_PIN) / 4.0;
216     float currentEgg = analogRead(EGG_PIN) / 4.0;
217     if (currentFlex > 10 && currentEgg > 10) {
218         flexSum += currentFlex;
219         eggSum += currentEgg;
220         validSamples++;
221     }
222     delay(100);
223 }
224 if (validSamples < 10) {
225     Serial.println(F("ERR: _Sensor_fault_during_calibration."));
226     currentState = STATE_ERROR;
227     return;
228 }
229 flexZero = flexSum / (float)validSamples;
230 eggSmooth = eggSum / (float)validSamples;
231 eggBaseline = eggSmooth;
232 initializeDSPBuffers(flexZero);
233 if (mpu.begin()) {
234     float xSum = 0, ySum = 0, zSum = 0;
235     for (int i = 0; i < 50; i++) {
236         sensors_event_t a, g, temp;
237         mpu.getEvent(&a, &g, &temp);
238         xSum += atan2(a.acceleration.y, a.acceleration.z) * 180.0 / PI;
239         ySum += atan2(-a.acceleration.x, sqrt(a.acceleration.y * a.acceleration.y
                + a.acceleration.z * a.acceleration.z)) * 180.0 / PI;
240         zSum += atan2(a.acceleration.x, a.acceleration.y) * 180.0 / PI;
241         delay(10);
242     }
243     baselineX = xSum / 50.0;
244     baselineY = ySum / 50.0;
245     baselineZ = zSum / 50.0;
246     calibratedIMU = true;
247     Serial.println(F("IMU_Calibration:_SUCCESS"));
248 } else {
249     calibratedIMU = false;
250     Serial.println(F("IMU_Calibration:_FAILED_(Using_Flex_Fallback)"));
251 }
252 triggerBeep(200, 100); triggerBeep(200, 0);
253 Serial.println(F("Calibration_Complete._Entering_Monitoring_Mode."));
254 currentState = STATE_MONITORING;
255 }
256
257 // 9. CORE ALERT LOGIC
258 void evaluateSystemAlerts() {

```

```

259 AlertType previousAlert = activeAlert;
260 activeAlert = ALERT_NONE;
261 if (abs(lumbarX) > LUMBAR_X_MAX || abs(lumbarY) > LUMBAR_Y_MAX || abs(lumbarZ)
    > LUMBAR_Z_MAX) {
262     activeAlert = ALERT_POSTURE_LUMBAR;
263 }
264 else if (abs(flexAngle) > FLEX_ANGLE_MAX) {
265     activeAlert = ALERT_POSTURE_FLEX;
266 }
267 else if ((respRateBPM < RESP_RATE_MIN && respRateBPM > 0) || respRateBPM >
    RESP_RATE_MAX) {
268     activeAlert = ALERT_RESPIRATORY;
269 }
270 else if (stomachActivity > STOMACH_ACTIVITY_MAX) {
271     activeAlert = ALERT_GASTRIC;
272 }
273 if (activeAlert != ALERT_NONE) {
274     manageAlertBuzzer(activeAlert);
275     if (activeAlert != previousAlert) {
276         Serial.print(F("\n>>>_CRITICAL_ALERT_DETECTED:_Code_"));
277         Serial.println(activeAlert);
278         SerialBT.print(F("MSG:_ALERT_"));
279         switch(activeAlert) {
280             case ALERT_POSTURE_LUMBAR: SerialBT.println("LUMBAR"); break;
281             case ALERT_POSTURE_FLEX: SerialBT.println("FLEX"); break;
282             case ALERT_RESPIRATORY: SerialBT.println("RESPIRATORY"); break;
283             case ALERT_GASTRIC: SerialBT.println("GASTRIC"); break;
284             default: break;
285         }
286     }
287 } else if (previousAlert != ALERT_NONE) {
288     digitalWrite(BUZZER_PIN, LOW);
289     Serial.println(F(">>>_Alert_Cleared._Back_to_Normal_"));
290     SerialBT.println(F("MSG:_NORMAL_"));
291 }
292 }
293
294 // 10. MAIN ARDUINO SETUP
295 void setup() {
296     Serial.begin(SERIAL_BAUD_RATE);
297     SerialBT.begin("GutAngle_Belt");
298     setupBuzzer();
299     triggerBeep(200, 100); triggerBeep(200, 0);
300     Serial.println(F("====="));
301     Serial.println(F("_GUTANGLE_BIOSIGNAL_ACQUISITION_SYSTEM_"));
302     Serial.println(F("====="));
303     Serial.println(F("Bootng_sequence_initiated..."));
304     if (!mpu.begin()) {
305         Serial.println(F("WRN:_MPU6050_unavailable._Check_I2C_wiring_"));

```

```

306     triggerBeep(100, 50); triggerBeep(100, 50); triggerBeep(500, 0);
307 } else {
308     mpu.setAccelerometerRange(MPU6050_RANGE_8_G);
309     mpu.setGyroRange(MPU6050_RANGE_500_DEG);
310     mpu.setFilterBandwidth(MPU6050_BAND_21_HZ);
311 }
312 currentState = STATE_CALIBRATING;
313 }
314
315 // 11. MAIN PROCESSING LOOP
316 void loop() {
317     unsigned long now = millis();
318     handleBluetoothCommands();
319     switch (currentState) {
320         case STATE_CALIBRATING:
321             runCalibration();
322             break;
323         case STATE_ERROR:
324             manageAlertBuzzer(ALERT_HARDWARE_FAIL);
325             delay(1000);
326             break;
327         case STATE_MONITORING:
328             if (now - lastSampleTime >= SAMPLE_RATE_MS) {
329                 lastSampleTime = now;
330                 currentData.eggRaw = analogRead(EGG_PIN) / 4.0;
331                 currentData.flexRaw = analogRead(FLEX_PIN) / 4.0;
332                 processEGGSignal(currentData.eggRaw);
333                 calculateRespirationRate();
334                 smoothedFlex = getMovingAverageFlex(currentData.flexRaw);
335                 float flexDiff = flexZero - smoothedFlex;
336                 if (abs(flexDiff) < 1.5) flexDiff = 0.0;
337                 flexAngle = flexDiff * 0.45;
338                 if (calibratedIMU) {
339                     sensors_event_t a, g, temp;
340                     mpu.getEvent(&a, &g, &temp);
341                     float rawX = atan2(a.acceleration.y, a.acceleration.z) * 180.0 / PI;
342                     float rawY = atan2(-a.acceleration.x, sqrt(a.acceleration.y * a.
                        acceleration.y + a.acceleration.z * a.acceleration.z)) * 180.0 /
                        PI;
343                     float rawZ = atan2(a.acceleration.x, a.acceleration.y) * 180.0 / PI;
344                     motionVariance = abs(g.gyro.x) + abs(g.gyro.y) + abs(g.gyro.z);
345                     float normX = rawX - baselineX;
346                     float normY = rawY - baselineY;
347                     float normZ = rawZ - baselineZ;
348                     lumbarX = (lumbarX * 0.8) + (normX * 0.2);
349                     lumbarY = (lumbarY * 0.8) + (normY * 0.2);
350                     lumbarZ = (lumbarZ * 0.8) + (normZ * 0.2);
351                 } else {
352                     lumbarX = flexAngle * 0.3;

```

```

353         lumbarY = flexAngle;
354         lumbarZ = 0.0;
355     }
356     evaluateSystemAlerts();
357     Serial.print(F("S:")); Serial.print(stomachActivity, 0);
358     Serial.print(F("F:")); Serial.print(flexAngle, 0);
359     Serial.print(F("X:")); Serial.print(lumbarX, 0);
360     Serial.print(F("Y:")); Serial.print(lumbarY, 0);
361     Serial.print(F("Z:")); Serial.print(lumbarZ, 0);
362     Serial.print(F("R:")); Serial.print(respRateBPM, 0);
363     if (activeAlert != ALERT_NONE) Serial.print(F("!")); else Serial.print(
        F("*"));
364     Serial.println();
365     SerialBT.print(stomachActivity, 0); SerialBT.print(",");
366     SerialBT.print(flexAngle, 0); SerialBT.print(",");
367     SerialBT.print(lumbarX, 0); SerialBT.print(",");
368     SerialBT.print(lumbarY, 0); SerialBT.print(",");
369     SerialBT.print(lumbarZ, 0); SerialBT.print(",");
370     SerialBT.println(respRateBPM, 0);
371 }
372 break;
373 }
374 }

```



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